Formally Verified PCP Constructions

Serhat Emre Coban   Davide Mazzali   Khanh Nguyen
Vincent Palma   Yanting Teng   Thomas Vidick   Yuxi Zheng

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Chapter 1

BLR for general finite fields

1.1 BLR over General Finite Fields

Throughout this section, let $q = p^s$ for a prime p and $s \in \mathbb{N}$ with $s \geq 1$. All vectors are in $\mathbb{F}_q^n$, and all expectations are uniform over the indicated finite sets. Let $\omega_p = e^{2\pi i/p}$.

Definition 1.1.1 (Field trace)**.** *The field trace* $\mathrm{Tr} : \mathbb{F}_q \to \mathbb{F}_p$ *is*

$$\mathrm{Tr}(z) := \sum_{i=0}^{s-1} z^{p^i}.$$

Definition 1.1.2 (Base additive character)**.** *The base additive character* $\psi : \mathbb{F}_q \to \mathbb{C}$ *is*

$$\psi(z) := \omega_p^{\mathrm{Tr}(z)}.$$

Definition 1.1.3 (Additive characters)**.** *For* $\alpha \in \mathbb{F}_q^n$*, define the additive character* $\chi_\alpha : \mathbb{F}_q^n \to \mathbb{C}$ *by*

$$\chi_\alpha(x) := \psi(\langle \alpha, x \rangle) = \omega_p^{\mathrm{Tr}(\langle \alpha, x \rangle)},$$

where $\langle \alpha, x \rangle = \sum_{i=1}^n \alpha_i x_i \in \mathbb{F}_q$*.*

Definition 1.1.4 (Expectation)**.** *For a function* $h : \mathbb{F}_q^n \to \mathbb{C}$*, define its uniform expectation by*

$$\mathbb{E}_{x \in \mathbb{F}_q^n}[h(x)] := \frac{1}{q^n} \sum_{x \in \mathbb{F}_q^n} h(x).$$

Definition 1.1.5 (Inner product)**.** *For functions* $h_1, h_2 : \mathbb{F}_q^n \to \mathbb{C}$*, define*

$$\langle h_1, h_2 \rangle := \mathbb{E}_{x \in \mathbb{F}_q^n}\left[h_1(x)\overline{h_2(x)}\right].$$

Definition 1.1.6 (L_2 norm)**.** *For a function* $h : \mathbb{F}_q^n \to \mathbb{C}$*, define its* L_2 *norm by*

$$\|h\|_2 := \left(\mathbb{E}_{x \in \mathbb{F}_q^n}\left[|h(x)|^2\right]\right)^{1/2} = \sqrt{\langle h, h \rangle}.$$

Definition 1.1.7 (Fourier coefficient)**.** *For* $h : \mathbb{F}_q^n \to \mathbb{C}$ *and* $\alpha \in \mathbb{F}_q^n$*, define*

$$\hat{h}(\alpha) := \langle h, \chi_\alpha \rangle = \mathbb{E}_{x \in \mathbb{F}_q^n}\left[h(x)\overline{\chi_\alpha(x)}\right].$$

Lemma 1.1.8 (Characters are orthonormal)**.** *For every* $\alpha, \beta \in \mathbb{F}_q^n$,

$$\langle \chi_\alpha, \chi_\beta \rangle = \begin{cases} 1 & \text{if } \alpha = \beta, \\ 0 & \text{if } \alpha \neq \beta. \end{cases}$$

Proof. First observe that for every $\alpha, \beta \in \mathbb{F}_q^n$ and every $x \in \mathbb{F}_q^n$,

$$\chi_\alpha(x)\overline{\chi_\beta(x)} = \omega_p^{\mathrm{Tr}(\langle \alpha, x \rangle)} \omega_p^{-\mathrm{Tr}(\langle \beta, x \rangle)} = \omega_p^{\mathrm{Tr}(\langle \alpha - \beta, x \rangle)} = \chi_{\alpha-\beta}(x),$$

where we used the $\mathbb{F}_p$-linearity of the trace map.

If $\alpha = \beta$, then $\alpha - \beta = 0$, so $\chi_{\alpha-\beta}(x) = 1$ for every $x \in \mathbb{F}_q^n$. Hence

$$\mathbb{E}_{x \in \mathbb{F}_q^n} \left[\chi_\alpha(x)\overline{\chi_\beta(x)} \right] = \mathbb{E}_{x \in \mathbb{F}_q^n}[1] = 1.$$

Now suppose $\alpha \neq \beta$. Let $\gamma := \alpha - \beta$. Then $\gamma \neq 0$, so there exists an index j such that $\gamma_j \neq 0$. Fix all coordinates of x except x_j. For the fixed remaining coordinates, the map

$$x_j \mapsto \langle \gamma, x \rangle$$

has the form

$$x_j \mapsto \gamma_j x_j + c$$

for some constant $c \in \mathbb{F}_q$. Since $\gamma_j \neq 0$, multiplication by γ_j is a bijection on $\mathbb{F}_q$, and therefore the affine map $x_j \mapsto \gamma_j x_j + c$ is also a bijection on $\mathbb{F}_q$. Thus, as x_j ranges uniformly over $\mathbb{F}_q$, the value $\langle \gamma, x \rangle$ also ranges uniformly over $\mathbb{F}_q$. Averaging first over x_j and then over the remaining coordinates gives

$$\mathbb{E}_{x \in \mathbb{F}_q^n} \left[\chi_\gamma(x) \right] = \mathbb{E}_{t \in \mathbb{F}_q} \left[\omega_p^{\mathrm{Tr}(t)} \right].$$

It remains to show that this last average is zero. The trace map $\mathrm{Tr} : \mathbb{F}_q \to \mathbb{F}_p$ is a nonzero $\mathbb{F}_p$-linear map, so every fiber of Tr has the same cardinality. Since $|\mathbb{F}_q| = q$ and $|\mathbb{F}_p| = p$, each fiber has cardinality q/p. Therefore

$$\begin{aligned} \mathbb{E}_{t \in \mathbb{F}_q} \left[\omega_p^{\mathrm{Tr}(t)} \right] &= \frac{1}{q} \sum_{t \in \mathbb{F}_q} \omega_p^{\mathrm{Tr}(t)} \\ &= \frac{1}{q} \sum_{r \in \mathbb{F}_p} \sum_{\substack{t \in \mathbb{F}_q \\ \mathrm{Tr}(t) = r}} \omega_p^r \\ &= \frac{1}{q} \sum_{r \in \mathbb{F}_p} \frac{q}{p} \omega_p^r \\ &= \frac{1}{p} \sum_{r \in \mathbb{F}_p} \omega_p^r \\ &= 0, \end{aligned}$$

because the sum of all p-th roots of unity is zero. Hence, if $\alpha \neq \beta$, then

$$\mathbb{E}_{x \in \mathbb{F}_q^n} \left[\chi_\alpha(x)\overline{\chi_\beta(x)} \right] = 0.$$

Combining the two cases proves the stated orthogonality relation. The equivalent formulation follows by taking $\beta = 0$. $\square$

Lemma 1.1.9 (Fourier inversion)**.** *For every function $h : \mathbb{F}_q^n \to \mathbb{C}$ we have*

$$h(x) = \sum_{\alpha \in \mathbb{F}_q^n} \hat{h}(\alpha)\chi_\alpha(x) \quad \textit{for every } x \in \mathbb{F}_q^n.$$

Proof. By Theorem 1.1.8, the additive characters

$$\{\chi_\alpha : \alpha \in \mathbb{F}_q^n\}$$

are orthonormal with respect to the inner product

$$\langle h_1, h_2 \rangle = \mathbb{E}_{x \in \mathbb{F}_q^n}\left[h_1(x)\overline{h_2(x)}\right].$$

Moreover, these characters span the vector space of all complex-valued functions on $\mathbb{F}_q^n$. Indeed, there are exactly q^n characters, and the space of functions $\mathbb{F}_q^n \to \mathbb{C}$ has dimension q^n. Since the characters are nonzero and pairwise orthogonal, they are linearly independent. Hence they form an orthonormal basis.

Therefore every function $h : \mathbb{F}_q^n \to \mathbb{C}$ has a unique expansion in this basis:

$$h = \sum_{\alpha \in \mathbb{F}_q^n} c_\alpha \chi_\alpha$$

for some coefficients $c_\alpha \in \mathbb{C}$. Taking the inner product of both sides with χ_β, we obtain

$$\langle h, \chi_\beta \rangle = \sum_{\alpha \in \mathbb{F}_q^n} c_\alpha \langle \chi_\alpha, \chi_\beta \rangle.$$

By orthonormality, all terms in the sum vanish except the term $\alpha = \beta$, and $\langle \chi_\beta, \chi_\beta \rangle = 1$. Hence

$$\langle h, \chi_\beta \rangle = c_\beta.$$

By definition of the Fourier coefficient,

$$\hat{h}(\beta) = \langle h, \chi_\beta \rangle.$$

Thus $c_\beta = \hat{h}(\beta)$ for every $\beta \in \mathbb{F}_q^n$. Therefore

$$h = \sum_{\alpha \in \mathbb{F}_q^n} \hat{h}(\alpha)\chi_\alpha.$$

Evaluating this identity at $x \in \mathbb{F}_q^n$ gives the Fourier inversion formula

$$h(x) = \sum_{\alpha \in \mathbb{F}_q^n} \hat{h}(\alpha)\chi_\alpha(x).$$

□

Lemma 1.1.10 (Plancherel identity)**.** *For every pair of functions $f, g : \mathbb{F}_q^n \to \mathbb{C}$,*

$$\sum_{\alpha \in \mathbb{F}_q^n} \hat{f}(\alpha)\overline{\hat{g}(\alpha)} = \langle f, g \rangle.$$

Proof. By Theorem 1.1.9,

$$f = \sum_{\alpha \in \mathbb{F}_q^n} \hat{f}(\alpha)\chi_\alpha.$$

Taking the inner product with g and using linearity in the first argument gives

$$\langle f, g\rangle = \sum_{\alpha \in \mathbb{F}_q^n} \hat{f}(\alpha)\langle \chi_\alpha, g\rangle.$$

Since

$$\langle \chi_\alpha, g\rangle = \overline{\langle g, \chi_\alpha\rangle} = \overline{\hat{g}(\alpha)},$$

we obtain

$$\langle f, g\rangle = \sum_{\alpha \in \mathbb{F}_q^n} \hat{f}(\alpha)\overline{\hat{g}(\alpha)}.$$

$\square$

Lemma 1.1.11 (Parseval identity)**.** *For every function $h : \mathbb{F}_q^n \to \mathbb{C}$,*

$$\sum_{\alpha \in \mathbb{F}_q^n} \left|\hat{h}(\alpha)\right|^2 = \|h\|_2^2.$$

Proof. Apply Theorem 1.1.10 with $f = g = h$:

$$\sum_{\alpha \in \mathbb{F}_q^n} \hat{h}(\alpha)\overline{\hat{h}(\alpha)} = \langle h, h\rangle.$$

The left-hand side is

$$\sum_{\alpha \in \mathbb{F}_q^n} \left|\hat{h}(\alpha)\right|^2,$$

and by the definitions of the inner product and the L_2 norm, the right-hand side is

$$\langle h, h\rangle = \mathbb{E}_{x \in \mathbb{F}_q^n}\left[|h(x)|^2\right] = \|h\|_2^2.$$

$\square$

Definition 1.1.12 (Fourier expansion of a finite-field-valued function)**.** *Let $f : \mathbb{F}_q^n \to \mathbb{F}_q$. For each $c \in \mathbb{F}_q^\times$, define*

$$\varphi_c(x) := \omega_p^{\mathrm{Tr}(cf(x))}.$$

The Fourier expansion of f is the family

$$\{\varphi_c\}_{c \in \mathbb{F}_q^\times}.$$

Lemma 1.1.13 (Character-sum indicator)**.** *For every $z \in \mathbb{F}_q$,*

$$\mathbb{1}[z = 0] = \frac{1}{q}\sum_{c \in \mathbb{F}_q} \omega_p^{\mathrm{Tr}(cz)}.$$

Consequently, for every $u, v \in \mathbb{F}_q$,

$$\mathbb{1}[u = v] = \frac{1}{q}\sum_{c \in \mathbb{F}_q} \omega_p^{\mathrm{Tr}(c(u-v))}.$$

Proof. Let

$$\psi(z) := \omega_p^{\mathrm{Tr}(z)}$$

be the standard additive character of $\mathbb{F}_q$. We prove that, for every $t \in \mathbb{F}_q$,

$$\mathbb{1}[t = 0] = \frac{1}{q} \sum_{c \in \mathbb{F}_q} \psi(ct).$$

The desired statement follows by taking $t = u - v$.

If $t = 0$, then $ct = 0$ for every $c \in \mathbb{F}_q$, so

$$\frac{1}{q} \sum_{c \in \mathbb{F}_q} \psi(ct) = \frac{1}{q} \sum_{c \in \mathbb{F}_q} 1 = 1.$$

Now suppose $t \neq 0$. Then multiplication by t is a bijection $\mathbb{F}_q \to \mathbb{F}_q$, so

$$\sum_{c \in \mathbb{F}_q} \psi(ct) = \sum_{z \in \mathbb{F}_q} \psi(z).$$

We claim that the latter sum is 0. Let

$$S := \sum_{z \in \mathbb{F}_q} \psi(z).$$

Since $\mathrm{Tr} : \mathbb{F}_q \to \mathbb{F}_p$ is a nonzero $\mathbb{F}_p$-linear map, it is surjective. Hence there exists $a \in \mathbb{F}_q$ such that $\mathrm{Tr}(a) = 1$. Using the bijection $z \mapsto z + a$, we get

$$S = \sum_{z \in \mathbb{F}_q} \psi(z + a) = \sum_{z \in \mathbb{F}_q} \omega_p^{\mathrm{Tr}(z+a)} = \sum_{z \in \mathbb{F}_q} \omega_p^{\mathrm{Tr}(z)} \omega_p^{\mathrm{Tr}(a)} = \omega_p S.$$

Since $\omega_p \neq 1$, this implies $S = 0$. Therefore, when $t \neq 0$,

$$\frac{1}{q} \sum_{c \in \mathbb{F}_q} \psi(ct) = 0.$$

Combining the two cases gives

$$\mathbb{1}[t = 0] = \frac{1}{q} \sum_{c \in \mathbb{F}_q} \omega_p^{\mathrm{Tr}(ct)}.$$

Substituting $t = u - v$, we obtain

$$\mathbb{1}[u = v] = \frac{1}{q} \sum_{c \in \mathbb{F}_q} \omega_p^{\mathrm{Tr}(c(u-v))}.$$

□

Definition 1.1.14 (Relative Hamming distance)**.** *For $f, g : \mathbb{F}_q^n \to \mathbb{F}_q$, define*

$$\delta(f, g) := \Pr_{x \in \mathbb{F}_q^n} \left[f(x) \neq g(x) \right].$$

Definition 1.1.15 (Linear functions). *For $\alpha \in \mathbb{F}_q^n$, define $\ell_\alpha : \mathbb{F}_q^n \to \mathbb{F}_q$ by*

$$\ell_\alpha(x) := \langle \alpha, x \rangle.$$

The set of linear functions is

$$\mathcal{LIN} := \left\{ \ell_\alpha \mid \alpha \in \mathbb{F}_q^n \right\}.$$

The distance from $f : \mathbb{F}_q^n \to \mathbb{F}_q$ to linearity is

$$\delta(f, \mathcal{LIN}) := \min_{\alpha \in \mathbb{F}_q^n} \delta(f, \ell_\alpha).$$

Lemma 1.1.16 (Well-separatedness of $\mathcal{LIN}$). *For any distincit $f, g \in \mathcal{LIN}$, one has $\delta(f, g) = 1 - 1/|\mathbb{F}|$.*

Proof. Let $a, b \in \mathbb{F}^n$ be the unique vectors such that $f(x) = \langle a, x \rangle$ and $g(x) = \langle b, x \rangle$ for each x. Let $c = a - b$, which must be a non-zero vector. In particular, there exists $i \in [n]$ such that $c_i \neq 0$. Take any permutation $\sigma : [n] \to [n]$ such that $\sigma(i) = n$. Then

$$\begin{aligned}
\delta(f, g) &= 1 - \Pr_{x \sim \mathbb{F}^n}[\langle c, x \rangle = 0] \\
&= 1 - \frac{1}{|\mathbb{F}|^{n-1}} \sum_{x_1, \ldots, x_{n-1} \in \mathbb{F}} \frac{1}{|\mathbb{F}|} \sum_{x_n \in \mathbb{F}} \mathbb{1}[c_i \cdot x_n = - \sum_{j \in [n] \setminus i} c_j \cdot x_{\sigma(j)}] \\
&= 1 - \frac{1}{|\mathbb{F}|^{n-1}} \sum_{x_1, \ldots, x_{n-1} \in \mathbb{F}} \frac{1}{|\mathbb{F}|} \sum_{x_n \in \mathbb{F}} \mathbb{1}[x_n = -c_i^{-1} \cdot \sum_{j \in [n] \setminus i} c_j \cdot x_{\sigma(j)}]
\end{aligned}$$

where the last equality follows because $c_i \neq 0$. Now note that $\mathbb{1}[x_n = -c_i^{-1} \cdot \sum_{j \in [n] \setminus i} c_j \cdot x_{\sigma(j)}]$ evaluates to 1 for exactly one value of x_n. Thus we continue from above and get

$$\delta(f, g) = 1 - \frac{1}{|\mathbb{F}|^{n-1}} \sum_{x_1, \ldots, x_{n-1} \in \mathbb{F}} \frac{1}{|\mathbb{F}|} = 1 - \frac{1}{|\mathbb{F}|}.$$

$\square$

Lemma 1.1.17 (Distance formula). *Let $f, g : \mathbb{F}_q^n \to \mathbb{F}_q$. Let $\{\varphi_c\}_{c \in \mathbb{F}_q^\times}$ and $\{\gamma_c\}_{c \in \mathbb{F}_q^\times}$ be their Fourier expansions. Then*

$$\delta(f, g) = 1 - \frac{1}{q} \left(1 + \sum_{c \in \mathbb{F}_q^\times} \langle \varphi_c, \gamma_c \rangle \right).$$

Equivalently,

$$\delta(f, g) = 1 - \frac{1}{q} \left(1 + \sum_{c \in \mathbb{F}_q^\times} \sum_{\alpha \in \mathbb{F}_q^n} \widehat{\varphi_c}(\alpha) \overline{\widehat{\gamma_c}(\alpha)} \right).$$

Proof. By Theorem 1.1.13,

$$\Pr_x[f(x) = g(x)] = \mathbb{E}_x \left[\frac{1}{q} \sum_{c \in \mathbb{F}_q} \omega_p^{\mathrm{Tr}(c(f(x) - g(x)))} \right].$$

The term $c = 0$ contributes $1/q$. For $c \neq 0$,

$$\omega_p^{\mathrm{Tr}(c(f(x) - g(x)))} = \varphi_c(x) \overline{\gamma_c(x)}.$$

Therefore

$$\Pr_x[f(x) = g(x)] = \frac{1}{q}\left(1 + \sum_{c \in \mathbb{F}_q^\times} \langle \varphi_c, \gamma_c \rangle\right),$$

which gives the first formula after using $\delta(f,g) = 1 - \Pr_x[f(x) = g(x)]$. The second formula follows from Fourier inversion and Parseval. $\square$

Lemma 1.1.18 (Fourier expansion of a linear function). *Fix $\alpha \in \mathbb{F}_q^n$ and $c \in \mathbb{F}_q^\times$. Let $\lambda_c(x) := \omega_p^{\mathrm{Tr}(c\ell_\alpha(x))}$. Then*

$$\lambda_c = \chi_{c\alpha}.$$

Equivalently, for every $\beta \in \mathbb{F}_q^n$,

$$\widehat{\lambda_c}(\beta) = \begin{cases} 1 & \text{if } \beta = c\alpha, \\ 0 & \text{if } \beta \neq c\alpha. \end{cases}$$

Proof. For every $x \in \mathbb{F}_q^n$,

$$\lambda_c(x) = \omega_p^{\mathrm{Tr}(c\langle \alpha, x\rangle)} = \omega_p^{\mathrm{Tr}(\langle c\alpha, x\rangle)} = \chi_{c\alpha}(x).$$

The statement about Fourier coefficients follows from Theorem 1.1.8. $\square$

Lemma 1.1.19 (Distance to linearity in Fourier form). *Let $f : \mathbb{F}_q^n \to \mathbb{F}_q$ have Fourier expansion $\{\varphi_c\}_{c \in \mathbb{F}_q^\times}$. Then*

$$\delta(f, \mathcal{LIN}) = 1 - \frac{1}{q}\left(1 + \max_{\alpha \in \mathbb{F}_q^n} \sum_{c \in \mathbb{F}_q^\times} \widehat{\varphi_c}(c\alpha)\right).$$

Moreover, for every $\alpha \in \mathbb{F}_q^n$, the quantity

$$S_\alpha := \sum_{c \in \mathbb{F}_q^\times} \widehat{\varphi_c}(c\alpha)$$

is real and satisfies $-1 \leq S_\alpha \leq q - 1$.

Proof. Apply Theorem 1.1.17 with $g = \ell_\alpha$ and use Theorem 1.1.18. This gives

$$\delta(f, \ell_\alpha) = 1 - \frac{1}{q}\left(1 + \sum_{c \in \mathbb{F}_q^\times} \widehat{\varphi_c}(c\alpha)\right).$$

Taking the minimum over α gives the claimed formula for $\delta(f, \mathcal{LIN})$. For fixed α, the same formula expresses S_α as $q(1 - \delta(f, \ell_\alpha)) - 1$. Hence S_α is real, and since $0 \leq \delta(f, \ell_\alpha) \leq 1$, we get $-1 \leq S_\alpha \leq q - 1$. $\square$

Definition 1.1.20 (Finite-field BLR test). *Given oracle access to $f : \mathbb{F}_q^n \to \mathbb{F}_q$, the finite-field BLR verifier $\mathsf{V}^f_{\mathsf{BLR}}$ samples $a, b \in \mathbb{F}_q^\times$ and $x, y \in \mathbb{F}_q^n$ uniformly and accepts iff*

$$af(x) + bf(y) = f(ax + by).$$

That is,

$$\mathsf{V}^f_{\mathsf{BLR}} = 1 \iff af(x) + bf(y) = f(ax + by).$$

Lemma 1.1.21 (Completeness)**.** *If $f \in \mathcal{LIN}$, then*

$$\Pr[\mathsf{V}_{\mathsf{BLR}}^{f} = 1] = 1.$$

Proof. If $f = \ell_\alpha$ for some $\alpha \in \mathbb{F}_q^n$, then for every $a, b \in \mathbb{F}_q^\times$ and $x, y \in \mathbb{F}_q^n$,

$$f(ax + by) = \langle \alpha, ax + by \rangle = a\langle \alpha, x \rangle + b\langle \alpha, y \rangle = af(x) + bf(y).$$

Thus the verifier accepts for every choice of randomness. □

Lemma 1.1.22 (Exact BLR acceptance formula)**.** *Let $f : \mathbb{F}_q^n \to \mathbb{F}_q$ have Fourier expansion $\{\varphi_c\}_{c \in \mathbb{F}_q^\times}$. For each $\alpha \in \mathbb{F}_q^n$, define*

$$S_\alpha := \sum_{c \in \mathbb{F}_q^\times} \widehat{\varphi_c}(c\alpha).$$

Then

$$\Pr[\mathsf{V}_{\mathsf{BLR}}^{f} = 1] = \frac{1}{q}\left(1 + \frac{1}{(q-1)^2} \sum_{\alpha \in \mathbb{F}_q^n} S_\alpha^3\right).$$

Proof. By Theorem 1.1.13, for fixed a, b, x, y,

$$\mathbb{1}[af(x) + bf(y) = f(ax + by)] = \frac{1}{q} \sum_{c \in \mathbb{F}_q} \omega_p^{\mathrm{Tr}(c(af(x)+bf(y)-f(ax+by)))}.$$

Taking expectation gives

$$\Pr[\mathsf{V}_{\mathsf{BLR}}^{f} = 1] = \frac{1}{q} + \frac{1}{q}\mathbb{E}_{a,b,x,y}\left[\sum_{c \in \mathbb{F}_q^\times} \varphi_{ca}(x)\varphi_{cb}(y)\varphi_{-c}(ax + by)\right].$$

Expand the three complex-valued functions into their Fourier expansions:

$$\begin{aligned}
&\mathbb{E}_{a,b,x,y}\left[\sum_{c \in \mathbb{F}_q^\times} \varphi_{ca}(x)\varphi_{cb}(y)\varphi_{-c}(ax + by)\right] \\
&= \mathbb{E}_{a,b}\left[\sum_{c \in \mathbb{F}_q^\times} \sum_{\alpha,\beta,\gamma \in \mathbb{F}_q^n} \widehat{\varphi_{ca}}(\alpha)\widehat{\varphi_{cb}}(\beta)\widehat{\varphi_{-c}}(\gamma)\mathbb{E}_{x,y}\left[\chi_\alpha(x)\chi_\beta(y)\chi_\gamma(ax + by)\right]\right].
\end{aligned}$$

Since

$$\chi_\gamma(ax + by) = \chi_{a\gamma}(x)\chi_{b\gamma}(y),$$

Theorem 1.1.8 implies that the inner expectation is 1 exactly when

$$\alpha + a\gamma = 0 \quad \text{and} \quad \beta + b\gamma = 0,$$

and is 0 otherwise. Therefore the last display equals

$$\mathbb{E}_{a,b}\left[\sum_{c \in \mathbb{F}_q^\times} \sum_{\alpha \in \mathbb{F}_q^n} \widehat{\varphi_{ca}}(\alpha)\widehat{\varphi_{cb}}(a^{-1}b\alpha)\widehat{\varphi_{-c}}(-a^{-1}\alpha)\right].$$

Substitute $\alpha = ca\eta$. Then this becomes

$$\mathbb{E}_{a,b}\left[\sum_{c \in \mathbb{F}_q^\times} \sum_{\eta \in \mathbb{F}_q^n} \widehat{\varphi_{ca}}(ca\eta)\widehat{\varphi_{cb}}(cb\eta)\widehat{\varphi_{-c}}(-c\eta)\right].$$

As a, b, c range over $\mathbb{F}_q^\times$, so do ca, cb, and $-c$. Hence this equals

$$\frac{1}{(q-1)^2} \sum_{\eta \in \mathbb{F}_q^n} \left(\sum_{d \in \mathbb{F}_q^\times} \widehat{\varphi_d}(d\eta) \right)^3 = \frac{1}{(q-1)^2} \sum_{\eta \in \mathbb{F}_q^n} S_\eta^3.$$

Substituting into the expression for $\Pr[\mathsf{V}_{\mathsf{BLR}}^f = 1]$ proves the claim. □

Lemma 1.1.23 (Second moment bound)**.** *With S_α as in Theorem 1.1.22,*

$$\sum_{\alpha \in \mathbb{F}_q^n} S_\alpha^2 \leq (q-1)^2.$$

Proof. Expanding the square gives

$$\sum_{\alpha \in \mathbb{F}_q^n} S_\alpha^2 = \sum_{a,b \in \mathbb{F}_q^\times} \sum_{\alpha \in \mathbb{F}_q^n} \widehat{\varphi_a}(a\alpha) \widehat{\varphi_b}(b\alpha).$$

For fixed $a, b \in \mathbb{F}_q^\times$, Cauchy–Schwarz and Theorem 1.1.11 give

$$\begin{aligned} \left| \sum_{\alpha \in \mathbb{F}_q^n} \widehat{\varphi_a}(a\alpha) \widehat{\varphi_b}(b\alpha) \right| &\leq \left(\sum_{\alpha \in \mathbb{F}_q^n} \left| \widehat{\varphi_a}(a\alpha) \right|^2 \right)^{1/2} \left(\sum_{\alpha \in \mathbb{F}_q^n} \left| \widehat{\varphi_b}(b\alpha) \right|^2 \right)^{1/2} \\ &= 1. \end{aligned}$$

The equality uses that multiplication by a and by b permutes $\mathbb{F}_q^n$, and that Parseval plus $|\varphi_a(x)| = |\varphi_b(x)| = 1$ for every x make both squared L_2 norms equal to 1. Summing over the $(q-1)^2$ choices of (a, b) gives the result. □

Theorem 1.1.24 (Soundness of the finite-field BLR test)**.** *For every function $f : \mathbb{F}_q^n \to \mathbb{F}_q$,*

$$\Pr[\mathsf{V}_{\mathsf{BLR}}^f = 0] \geq \delta(f, \mathcal{LIN}).$$

Equivalently,

$$\Pr[\mathsf{V}_{\mathsf{BLR}}^f = 1] \leq 1 - \delta(f, \mathcal{LIN}).$$

Proof. Let S_α be as in Theorem 1.1.22, and let

$$M := \max_{\alpha \in \mathbb{F}_q^n} S_\alpha.$$

By Theorem 1.1.19,

$$1 - \delta(f, \mathcal{LIN}) = \frac{1}{q}(1 + M).$$

By Theorem 1.1.19, each S_α is real and $S_\alpha \leq M$. Therefore $S_\alpha^3 \leq M S_\alpha^2$ for every α, because $S_\alpha^2 \geq 0$. Using Theorem 1.1.22 and Theorem 1.1.23,

$$\begin{aligned} \Pr[\mathsf{V}_{\mathsf{BLR}}^f = 1] &= \frac{1}{q} \left(1 + \frac{1}{(q-1)^2} \sum_{\alpha \in \mathbb{F}_q^n} S_\alpha^3 \right) \\ &\leq \frac{1}{q} \left(1 + \frac{M}{(q-1)^2} \sum_{\alpha \in \mathbb{F}_q^n} S_\alpha^2 \right) \\ &\leq \frac{1}{q}(1 + M) \\ &= 1 - \delta(f, \mathcal{LIN}). \end{aligned}$$

Taking complements gives $\Pr[\mathsf{V}_{\mathsf{BLR}}^{f} = 0] \geq \delta(f, \mathcal{LIN})$. $\square$

Theorem 1.1.25 (Counterexample to the proposed cubic lower bound)**.** *The inequality*

$$(1 - \delta(f, \mathcal{LIN}))^3 \leq \Pr[\mathsf{V}_{\mathsf{BLR}}^{f} = 1]$$

does not hold for all functions $f : \mathbb{F}_q^n \to \mathbb{F}_q$ with the finite-field BLR verifier from Theorem 1.1.20.

Proof. It suffices to give a concrete counterexample. Take $q = 2$ and $n = 1$, so the domain and codomain are both $\mathbb{F}_2$. Let

$$f : \mathbb{F}_2 \to \mathbb{F}_2, \qquad f(0) = 1, \quad f(1) = 1.$$

The homogeneous linear maps $\mathbb{F}_2 \to \mathbb{F}_2$ are exactly the zero map and the identity map. The function f agrees with the zero map at no point, and it agrees with the identity map only at the point 1. Hence

$$\delta(f, \mathcal{LIN}) = \frac{1}{2}, \qquad 1 - \delta(f, \mathcal{LIN}) = \frac{1}{2}.$$

For the verifier, the only possible nonzero field elements are $a = b = 1$. Thus the verifier accepts on a pair (x, y) exactly when

$$f(x) + f(y) = f(x + y).$$

Since f is constantly 1, the left-hand side is $1 + 1 = 0$ in $\mathbb{F}_2$, while the right-hand side is 1. Therefore the verifier rejects for every choice of x, y, and

$$\Pr[\mathsf{V}_{\mathsf{BLR}}^{f} = 1] = 0.$$

But

$$(1 - \delta(f, \mathcal{LIN}))^3 = \left(\frac{1}{2}\right)^3 = \frac{1}{8} > 0 = \Pr[\mathsf{V}_{\mathsf{BLR}}^{f} = 1].$$

This contradicts the proposed lower bound. $\square$

Theorem 1.1.26 (Lower bound from agreement with a linear function)**.** *Let $f : \mathbb{F}_q^n \to \mathbb{F}_q$. Suppose there exists $\ell \in \mathcal{LIN}$ such that*

$$\Pr_{x \in \mathbb{F}_q^n}[f(x) = \ell(x)] \geq 1 - \varepsilon.$$

Then

$$\Pr[\mathsf{V}_{\mathsf{BLR}}^{f} = 1] \geq 1 - 3\varepsilon + 2\varepsilon^2.$$

In particular,

$$\Pr[\mathsf{V}_{\mathsf{BLR}}^{f} = 1] \geq 1 - 3\delta(f, \mathcal{LIN}) + 2\delta(f, \mathcal{LIN})^2.$$

Proof. Let

$$G := \{z \in \mathbb{F}_q^n : f(z) = \ell(z)\}$$

be the agreement set, and let $B := \mathbb{F}_q^n \;\; G$ be the bad set. Write

$$\eta := \Pr_z[z \in B].$$

By assumption, $\eta \leq \varepsilon$.

The verifier samples $a, b \in \mathbb{F}_q^\times$ and $x, y \in \mathbb{F}_q^n$, and queries

$$x, \qquad y, \qquad ax + by.$$

For fixed $a, b \in \mathbb{F}_q^\times$, the random variables x, y, and $ax + by$ are each uniform on $\mathbb{F}_q^n$. Indeed, x and y are uniform by definition, and $ax+by$ is uniform because, after fixing a, b, and x, the map

$$y \mapsto ax + by$$

is a bijection of $\mathbb{F}_q^n$.

Therefore

$$\Pr[x \in B] = \Pr[y \in B] = \Pr[ax + by \in B] = \eta.$$

Moreover the three queried points are pairwise independent. For fixed $a, b \in \mathbb{F}_q^\times$, each of the maps

$$(x, y) \mapsto (x, y), \qquad (x, y) \mapsto (x, ax + by), \qquad (x, y) \mapsto (y, ax + by)$$

is a bijection from $\mathbb{F}_q^n \times \mathbb{F}_q^n$ to itself. Hence

$$\Pr[x \in B,\ y \in B] = \Pr[x \in B,\ ax + by \in B] = \Pr[y \in B,\ ax + by \in B] = \eta^2.$$

By inclusion–exclusion and the trivial bound

$$\Pr[x \in B,\ y \in B,\ ax + by \in B] \leq \Pr[x \in B,\ y \in B] = \eta^2,$$

we get

$$\Pr[x \in B \text{ or } y \in B \text{ or } ax + by \in B] \leq 3\eta - 3\eta^2 + \eta^2 = 3\eta - 2\eta^2.$$

On the complementary event, the verifier sees the same three values as it would see from ℓ:

$$f(x) = \ell(x), \qquad f(y) = \ell(y), \qquad f(ax + by) = \ell(ax + by).$$

Since ℓ is linear,

$$\ell(ax + by) = a\ell(x) + b\ell(y).$$

Thus

$$f(ax + by) = af(x) + bf(y),$$

so the verifier accepts. Hence the acceptance probability is at least the probability of the complementary event:

$$\Pr[\mathsf{V}^f_{\mathsf{BLR}} = 1] \geq 1 - 3\eta + 2\eta^2.$$

If $\varepsilon \geq 1/2$, then $1 - 3\varepsilon + 2\varepsilon^2 \leq 0$, so the desired bound is trivial. Otherwise $\varepsilon \leq 1/2$, and since $\eta \leq \varepsilon$, the function $1 - 3t + 2t^2$ is decreasing on the interval containing both η and ε. This gives

$$\Pr[\mathsf{V}^f_{\mathsf{BLR}} = 1] \geq 1 - 3\varepsilon + 2\varepsilon^2.$$

Taking ℓ to be a closest linear function gives the final statement with $\varepsilon = \delta(f, \mathcal{LIN})$. □

Theorem 1.1.27 (Boolean tightness for subspace error sets)**.** *The bound in Theorem 1.1.26 is tight for the Boolean BLR test for infinitely many values of ε. More precisely, let $q = 2$ and let $B \leq \mathbb{F}_2^n$ be a linear subspace of density $\varepsilon = 2^{-k}$. Define*

$$f : \mathbb{F}_2^n \to \mathbb{F}_2, \qquad f(x) = \begin{cases} 1 & \text{if } x \in B, \\ 0 & \text{if } x \notin B. \end{cases}$$

Then $\delta(f, \mathcal{LIN}) = \varepsilon$ and

$$\Pr[\mathsf{V}^f_{\mathsf{BLR}} = 1] = 1 - 3\varepsilon + 2\varepsilon^2.$$

Proof. The zero linear function disagrees with f exactly on B, so $\delta(f, \mathcal{LIN}) \leq \varepsilon$. If $m : \mathbb{F}_2^n \to \mathbb{F}_2$ is any nonzero linear function, then m is balanced. Thus m agrees with the indicator of B on at most half of the domain, whereas the zero linear function agrees on a $1-\varepsilon$ fraction of the domain. For $\varepsilon \leq 1/2$, this shows that the closest linear function is the zero function, so $\delta(f, \mathcal{LIN}) = \varepsilon$.

Since $q = 2$, the verifier has $a = b = 1$ and accepts iff

$$f(x) + f(y) = f(x + y).$$

Because B is a subspace, among the three points x, y, and $x + y$, the number lying in B is never exactly two. Therefore the verifier rejects exactly when at least one of the three queried points lies in B.

The three events

$$x \in B, \qquad y \in B, \qquad x + y \in B$$

each have probability ε, and each pair has probability ε^2. Since B is a subspace, the triple event also has probability ε^2: if $x, y \in B$, then $x + y \in B$. Hence

$$\Pr[\mathsf{V}_{\mathsf{BLR}}^f = 0] = 3\varepsilon - 3\varepsilon^2 + \varepsilon^2 = 3\varepsilon - 2\varepsilon^2.$$

Taking complements gives

$$\Pr[\mathsf{V}_{\mathsf{BLR}}^f = 1] = 1 - 3\varepsilon + 2\varepsilon^2.$$

□

Chapter 2

Gowers Hatami theorem

2.1 Preliminaries

Definition 2.1.1 (σ inner product.). *The σ inner product between matrices A and B is defined as*

$$\langle A, B \rangle_\sigma = \mathrm{Tr}(A^* B \sigma), \tag{2.1}$$

where $\sigma \geq 0$ and A^* denotes conjugate-transpose.

Definition 2.1.2 ((ϵ, σ)-representation.). *Given a finite group G, an integer $d \geq 1, \varepsilon \geq 0$, and a d-dimensional positive semidefinite matrix σ with trace 1 , an (ε, σ)-representation of G is a map $f : G \to U_d(\mathbb{C})$, to the $d \times d$ unitary matrices, such that*

$$\mathbb{E}_{x,y \in G} \Re \left(\left\langle f(x)^* f(y), f\left(x^{-1} y\right) \right\rangle_\sigma \right) \geq 1 - \varepsilon, \tag{2.2}$$

where $\Re$ denotes taking real part of the inner product.

Proposition 2.1.3. *Let G be a finite group, and let $f, f_2 : G \to U(n)$ be functions into the unitary matrices. Then*

$$\mathbb{E}_{x \in G} \| f(x) - f_2(x) \|_\sigma^2 \leq 2\varepsilon \quad \Longleftrightarrow \quad \mathbb{E}_{x \in G} \Re(\langle f(x), f_2(x) \rangle_\sigma) \geq 1 - \varepsilon.$$

Proof. For unitary matrices A, B, we have

$$\| A - B \|_\sigma^2 = 2 - 2\Re \langle A, B \rangle_\sigma. \tag{2.3}$$

Since $\mathbb{E}$ is linear, this proves the proposition. $\square$

TODO:

- Relation to classical BLR test

2.2 Representation theory

Definition 2.2.1 (Unitary Representation). *A* unitary representation *of G is a representation to unitary matrices $\rho : G \to U(V)$.*

Definition 2.2.2 (Regular representation)**.** *Let $R : G \to \mathbb{C}[G]$ be a regular representation where $\{|g\rangle, g \in G\}$ forms a basis of $\mathbb{C}[G]$. The representation is given by*

$$R(x)|g\rangle = |xg\rangle. \tag{2.4}$$

It is possible to decompose any representation of a finite group into direct sums over irreps.

Theorem 2.2.3 (Maschke's theorem)**.** *Let G be a finite group and let $\rho : G \to \mathrm{GL}(V)$ be a finite-dimensional representation over $\mathbb{C}$. Then ρ is completely reducible. In particular, there exists a decomposition*

$$\rho \cong \bigoplus_m r_m \, \rho_m, \tag{2.5}$$

where $\{\rho_m\}$ is a set of inequivalent irreducible representations of G, and $r_m \in \mathbb{Z}^+$ are the multiplicities.

Proof sketch. Maschke's theorem is partially formalized in `Mathlib.RepresentationTheory.Maschke` but incomplete. Jonathan's team is formalizing this. □

Proposition 2.2.4 (Decomposition of the regular representation)**.** *Let $R : G \to \mathrm{GL}(\mathbb{C}[G])$ be the regular representation of a finite group G. Then*

$$R \cong \bigoplus_{\rho \in \widehat{G}} d_\rho \, \rho, \tag{2.6}$$

where $\widehat{G}$ is a complete set of inequivalent irreducible representations of G, and $d_\rho = \dim(V_\rho)$ is the dimension of ρ.

Proof sketch. This can be proven using Maschke's theorem in 2.2.3 to decompose into irreps with multiplicity. Then use character orthogonality theorem in mathlib `FDRep.char_orthonormal` to show the multiplicity equals irrep dimension. A more formal statement is given by the Peter-Weyl theorem. □

Proposition 2.2.5 (Character of regular representation.)**.** *Let G be a finite group and $\sum_\rho$ denotes summing over the complete set of inequivalent irreps $\hat{G}$*

$$\sum_{\rho \in \hat{G}} d_\rho \, \mathrm{Tr}(\rho(x)) = |G| \delta_{x,e}. \tag{2.7}$$

Proof sketch. The proof uses decomposition of regular representation and definition of character. □

Definition 2.2.6 (Group fourier transform)**.** *Let $f : G \to U_d(\mathbb{C})$ be a map and $\rho : G \to U_{d_\rho}(\mathbb{C})$ be a unitary irrep. The fourier transform of f at the representation ρ is denoted by $\hat{f}$:*

$$\hat{f}(\rho) = \mathbb{E}_{x \in G} f(x) \otimes \overline{\rho(x)}, \tag{2.8}$$

where $\overline{\rho(x)}$ denotes complex conjugate of $\rho(x)$.

2.3 Gowers Hatami Theorem

Theorem 2.3.1 (Gowers-Hatami)**.** *Let G be a finite group, $\varepsilon \geq 0$, and $f : G \to U_d(\mathbb{C})$ an (ε, σ)-representation of G. Then there exists a $d' \geq d$, an isometry $V : \mathbb{C}^d \to \mathbb{C}^{d'}$, and a representation $g : G \to U_{d'}(\mathbb{C})$ such that*

$$\mathbb{E}_{x \in G} \left\| f(x) - V^* g(x) V \right\|_\sigma^2 \leq 2\varepsilon. \tag{2.9}$$

Proof sketch. The full proof can be found in the lecture notes Theorem 2.3 [?]. In particular, the proof is constructive. The isometry is defined as $V : \mathbb{C}^d \to \mathbb{C}^d \otimes \left(\oplus_\rho \mathbb{C}^{d_\rho} \otimes \mathbb{C}^{d_\rho}\right)$ by

$$Vu = \bigoplus_\rho d_\rho^{1/2} \sum_{i=1}^{d_\rho} \left(\hat{f}(\rho)\left(u \otimes e_i\right)\right) \otimes e_i. \tag{2.10}$$

And the exact representation is defined as

$$g(x) = \bigoplus_\rho \left(I_d \otimes I_{d_\rho} \otimes \rho(x)\right). \tag{2.11}$$

The proof requires the following two ingredients:

1. V is an isometry:
$$V^* V = \mathbb{I}_d \tag{2.12}$$

2. The isometry "averages" over group multiplication. For any $x \in G$,
$$V^* g(x) V = \mathbb{E}_z f(z)^* f(zx). \tag{2.13}$$

Both ingredients follow from the key identity over the regular representation in Eq. (2.7). □

Next, we will give a different proof that does not directly decompose into irreps but instead uses the action of the regular representation.

Lemma 2.3.2 (Gowers-Hatami-Correlation)**.** *Let G be a finite group, $\varepsilon \geq 0$, and $\rho : G \to U(\mathcal{H})$ an (ε, σ)-representation of G on space $\mathcal{H}$. Then there exists an isometry $V : \mathcal{H} \to \mathcal{H}'$ to a different space $\mathcal{H}'$, and a representation $\rho' : G \to \mathcal{H}'$ such that the average correlation between ρ and ρ' is:*

$$\mathbb{E}_{x \in G} \Re \langle f(x), V^* \rho'(x) V \rangle_\sigma \geq 1 - \varepsilon, \tag{2.14}$$

Proof. Here $\mathcal{H}' = L(G, \mathcal{H})$ is the space of functions $f : G \to \mathcal{H}$.

Let the isometry be defined by the action of the approximate representation on a state $u \in \mathcal{H}$ as

$$V : \quad \mathcal{H} \to L(G, \mathcal{H}), \tag{2.15}$$
$$V(u) : \quad G \to \mathcal{H}, \quad x \mapsto \rho(x)(u), \forall x \in G. \tag{2.16}$$

The adjoint map V^* is defined as the unique map such that for all $u \in \mathcal{H}$ and $g \in L(G, \mathcal{H})$,

$$\langle V(u), g \rangle_{L(G,\mathcal{H})} = \langle u, V^*(g) \rangle_{\mathcal{H}}. \tag{2.17}$$

In particular, V^* is explicitly given by the group average

$$V^* : \quad L(G, \mathcal{H}) \to \mathcal{H}, \tag{2.18}$$
$$V^*(f) \quad = \mathbb{E}_{x \in G} \left[\rho^*(x) f(x)\right]. \tag{2.19}$$

The exact representation is given by the right regular representation on $L(G, \mathcal{H})$:

$$\begin{aligned} \rho' : \qquad & G \to \mathrm{End}(L(G, \mathcal{H})), && (2.20)\\ (\rho'(x)f)(y) \quad & = f(yx), \quad \forall x, y \in G. && (2.21) \end{aligned}$$

V is an isometry because

$$V^*V(u) = \mathbb{E}_{x\in G}\left[\rho^*(x)\rho(x)(u)\right] = u. \tag{2.22}$$

The isometry "averages" over group multiplication because

$$\begin{aligned} V^*\rho'(x)V(u) \quad &= \quad \mathbb{E}_{y\in G}\left[\rho^*(y)(\rho'(x)\rho(y))(u)\right] \qquad (2.23)\\ &= \quad \mathbb{E}_{y\in G}\left[\rho^*(y)(\rho(yx)(u))\right]. \end{aligned}$$

Finally, using the isometry average property in Eq. 2.23 and the definition of (ε, σ)-representation, we have

$$\begin{aligned} & \mathbb{E}_{x\in G}\Re\langle\rho(x), V^*\rho'(x)V\rangle_\sigma, \qquad (2.24)\\ = \quad & \mathbb{E}_{x,y\in G}\Re\langle\rho(x), \rho^*(y)\rho(yx)\rangle_\sigma,\\ \geq \quad & 1 - \varepsilon. \end{aligned}$$

□

Theorem 2.3.3 (Gowers-Hatami-2)**.** *Let G be a finite group, $\varepsilon \geq 0$, and $\rho : G \to U(\mathcal{H})$ an (ε, σ)-representation of G on space $\mathcal{H} = \mathbb{C}^d$. Then there exists an isometry $V : \mathcal{H} \to \mathcal{H}'$ to a different space $\mathcal{H}'$, and a representation $\rho' : G \to \mathcal{H}'$ such that*

$$\mathbb{E}_{x\in G}\left\|\rho(x) - V^*\rho'(x)V\right\|_\sigma^2 \leq 2\varepsilon. \tag{2.25}$$

Proof. A different proof of the Gowers-Hatami theorem is given in M. de la Salle's notes [**?**].

The rest of the proof follows by expanding the σ-norm, bounding the norm of $\mathbb{E}_{x\in G}V^*\rho'(x)V$ and applying the definition of (ε, σ)-representation. In particular, we have

$$\begin{aligned} & \mathbb{E}_{x\in G}\left\|\rho(x) - V^*\rho'(x)V\right\|_\sigma^2 \qquad (2.26)\\ = \quad & \mathbb{E}_{x\in G}\|\rho(x)\|_\sigma^2 + \mathbb{E}_{x\in G}\|V^*\rho'(x)V\|_\sigma^2 - 2\mathbb{E}_{x\in G}\Re\langle\rho(x), V^*\rho'(x)V\rangle_\sigma \end{aligned}$$

Because ρ is unitary, $\mathbb{E}_{x\in G}\|\rho(x)\|_\sigma^2 = 1$. For the second term, since ρ' is unitary and V is an isometry,

$$\begin{aligned} & \mathbb{E}_{x\in G}\|V^*\rho'(x)V\|_\sigma^2 \qquad (2.27)\\ = \quad & \mathbb{E}_{x\in G}\mathrm{Tr}\left[V^*\rho'^*(x)VV^*\rho'(x)V\sigma\right]\\ = \quad & \mathbb{E}_{x,y,z\in G}\mathrm{Tr}\left[\rho^*(yx)\rho(y)\rho^*(z)\rho(zx)\sigma\right] \end{aligned}$$

Because ρ is unitary the product $U(x, y, z) := \rho^*(yx)\rho(y)\rho^*(z)\rho(zx)$ is again a unitary operator, the above expression is upper bounded by Holder's inequality as

$$\begin{aligned} & \mathbb{E}_{x,y,z\in G}\mathrm{Tr}\left[U(x, y, z)\sigma\right] \qquad (2.28)\\ \leq \quad & \mathbb{E}_{x,y,z\in G}\|U(x, y, z)\|_\infty\|\sigma\|_1 = 1. \end{aligned}$$

Therefore using the bounds in Eq. 2.28 and Eq. 2.24 by applying Lemma 2.3.2, we have

$$\mathbb{E}_{x\in G}\left\|\rho(x) - V^*\rho'(x)V\right\|_\sigma^2 \leq 2 - 2(1 - \varepsilon) = 2\varepsilon. \tag{2.29}$$

□

2.4 Gowers-Hatami for BLR linearity test over $\mathbb{F}_2$

In classical BLR, the mapping is from a field $\mathbb{F}_2^n$. First let us relate the BLR test to the notion of approximate representation.

Proposition 2.4.1 (BLR test induces an approximate representation). *Let $f : \mathbb{Z}_2^n \to \mathbb{Z}_2$ satisfy*

$$\Pr_{x,y}[f(x) + f(y) = f(x+y)] \geq 1 - \epsilon.$$

Define $F : \mathbb{Z}_2^n \to \{\pm 1\}$ by $F(x) = (-1)^{f(x)}$. Then

$$\mathbb{E}_{x,y}\big|F(x)F(y) - F(x+y)\big|^2 \leq 4\epsilon,$$

so F is an $(2\epsilon, 1)$-approximate representation.

Proof. First observe that for all $x, y \in G$,

$$F(x)F(y) = (-1)^{f(x)+f(y)}.$$

Hence

$$F(x)F(y) = F(x+y) \quad \Longleftrightarrow \quad f(x) + f(y) = f(x+y) \pmod 2.$$

Define the indicator of BLR success:

$$\mathbf{1}_{\text{good}}(x, y) := \mathbf{1}[f(x) + f(y) = f(x+y)].$$

Then

$$\Pr_{x,y}[\mathbf{1}_{\text{good}} = 1] \geq 1 - \epsilon.$$

Now compute the squared error. Since $F(x) \in \{\pm 1\}$,

$$|F(x)F(y) - F(x+y)|^2 = \begin{cases} 0 & \text{if } F(x)F(y) = F(x+y), \\ 4 & \text{otherwise.} \end{cases}$$

Taking expectation over (x, y),

$$\begin{aligned} \mathbb{E}_{x,y}|F(x)F(y) - F(x+y)|^2 &= 4 \Pr_{x,y}[F(x)F(y) \neq F(x+y)] \\ &= 4 \Pr_{x,y}[f(x) + f(y) \neq f(x+y)] \\ &\leq 4\epsilon. \end{aligned}$$

□

The Gowers-Hatami theorem can be applied to show that if a function $f : \mathbb{Z}_2^n \to \mathbb{F}_2$ passes the BLR linearity test with high probability, then there exists a linear function $\ell : \mathbb{F}_2^n \to \mathbb{F}_2$ that is close to f in terms of Hamming distance $\delta(f, \ell)$ defined in Def. 1.1.14. To do this, we first need to apply the Gowers-Hatami theorem to the approximate representation $F(x) = (-1)^{f(x)}$ given by Proposition 2.4.1.

Corollary 2.4.2 (Gowers-Hatami Correlation for BLR). *Let $F : \mathbb{Z}_2^n \to U(1)$ be an (ϵ, σ) approximate representation. Then there exists a unit vector v and an exact representation G such that*

$$\mathbb{E}_{x \in \mathbb{Z}_2^n} \Re\langle F(x), \langle v, G(x)v\rangle\rangle \geq 1 - \epsilon. \tag{2.30}$$

Next, we would like to conclude from the above corollary that there exists a linear function $\ell : \mathbb{Z}_2^n \to \mathbb{Z}_2$ such that $\Pr_x[f(x) \neq \ell(x)] \leq \epsilon/2$. To do this, we need to use a fact that about the regular representation of $\mathbb{Z}_2^n$.

Fact 2.4.3 (Regular representation of $\mathbb{Z}_2^n$)**.** *Let $\rho(a)$ act on $L = \{f : \mathbb{Z}_2^n \to \mathbb{C}\}$ by $(\rho(a)f)(x) = f(x+a)$. Then in the Fourier basis $\{|S\rangle\}$, we have*

$$\rho(a) = \sum_{S \subseteq [n]} \chi_S(a)|S\rangle\langle S|.$$

Lemma 2.4.4 (Gowers-Hatami implies near linearity)**.** *Let $F : \mathbb{Z}_2^n \to \mathbb{C}$ be an $(\epsilon, 1)$ approximate representation of $\mathbb{Z}_2^n$ from the BLR test such that $F(x) = (-1)^{f(x)}$ for some function $f : \mathbb{Z}_2^n \to \mathbb{Z}_2$. Then there exists a linear function $\ell : \mathbb{F}_2^n \to \mathbb{F}_2$ such that the distance between f and ℓ is bounded*

$$\delta(f, \ell) \leq \epsilon/2. \tag{2.31}$$

Proof. By Corollary 2.4.2, there exists a representation G and a unit vector v such that

$$\mathbb{E}_x \Re\langle F(x), \langle v, G(x)v\rangle\rangle \geq 1 - \epsilon.$$

Using the diagonal form of $G(x)$ in the character basis,

$$G(x) = \sum_S \chi_S(x)|S\rangle\langle S|.$$

The inner product condition can be rewritten as

$$\sum_S p_S \mathbb{E}_x \Re\langle F(x), \chi_S(x)\rangle \geq 1 - \epsilon,$$

where $p_S := |\langle v|S\rangle|^2$ and $\sum_S p_S = 1$. Therefore,

$$\sum_S p_S \widehat{F}(S) \geq 1 - \epsilon,$$

where $\widehat{F}(S) := \mathbb{E}_x[F(x)\chi_S(x)]$ is the real-valued Fourier coefficient of F at S.

Since $(p_S)_S$ is a probability distribution, there exists some S^* such that

$$\widehat{F}(S^*) \geq 1 - \epsilon.$$

This means that there is a large Fourier coefficient at S^*. Define $\ell(x) = x \cdot S^*$. Then $\chi_{S^*}(x) = (-1)^{\ell(x)}$, and

$$\mathbb{E}_x[F(x)\chi_{S^*}(x)] = 1 - 2\Pr_x[f(x) \neq \ell(x)].$$

Hence

$$1 - 2\delta(f, \ell) \geq 1 - \epsilon,$$

which implies

$$\delta(f, \ell) \leq \epsilon/2.$$

□

Proposition 2.4.5 (Equivalence for BLR test)**.** *Let $\Pr[V_{BLR}^f = 0]$ denote the probability that the BLR verifier rejects a function $f : \mathbb{F}_2^n \to \mathbb{F}_2$. Let $\delta(f, \mathcal{LIN})$ denote the distance between f and the closest linear function in $\mathcal{LIN}$. Then the following statements are equivalent:*

- *If* $\Pr[V_{BLR}^f = 0] \le \epsilon$, *then* $\delta(f, \mathcal{LIN}) \le \epsilon$.
- $\Pr[V_{BLR}^f = 0] \ge \delta(f, \mathcal{LIN})$.

Now we provide the proof of Soundness of BLR test using the Gowers-Hatami theorem.

Theorem 2.4.6 (Soundness of the Z_2 BLR)**.** *For every function* $f : \mathbb{F}_2^n \to \mathbb{F}_2$,

$$\Pr[V_{BLR}^f = 0] \ge \delta(f, \mathcal{LIN}).$$

Equivalently,

$$\Pr[V_{BLR}^f = 1] \le 1 - \delta(f, \mathcal{LIN}).$$

Proof. Let $F : \mathbb{Z}_2^n \to \{-1, 1\}$ be defined by $F(x) = (-1)^{f(x)}$, as given by Proposition .2.4.1. F is an $(2\epsilon, 1)$ approximate representation of $\mathbb{Z}_2^n$. Applying Gowers-Hatami theorem by Lemma. 2.4.4, then there exists a linear function ℓ such that $\delta(f, \ell) \le \epsilon$ so that $\delta(f, \mathcal{LIN}) \le \epsilon$.

This proves the implication $\Pr[V_{\mathrm{BLR}}^f = 0] \le \epsilon \Rightarrow \delta(f, \mathcal{LIN}) \le \epsilon$, which is equivalent to

$$\Pr[V_{\mathrm{BLR}}^f = 0] \ge \delta(f, \mathcal{LIN}),$$

by Proposition 2.4.5. □

2.5 Gowers-Hatami for BLR linearity test over $\mathbb{F}_q$

TODO: add Trace for prime powers

Assume $q = p$ is a prime, $n \ge 1$, and we set

$$X := \mathbb{F}_q^n, \qquad A := \mathbb{F}_q^\times, \qquad \omega := e^{2\pi i/p}.$$

Definition 2.5.1 (BLR verifier and distance)**.** *Given* $f : X \to \mathbb{F}_q$, *the verifier* $\mathsf{V}_{\mathrm{BLR}}^f$ *samples* $a, b \in A$ *and* $x, y \in X$ *uniformly and accepts iff* $af(x) + bf(y) = f(ax + by)$. *Let the test rejection probability be bounded by*

$$\Pr_{a,b,x,y}\big[af(x) + bf(y) \ne f(ax + by)\big] \le \varepsilon.$$

For $\alpha \in X$, *write* $\ell_\alpha(x) := \langle \alpha, x \rangle$, $\delta_\alpha := \Pr_x[f(x) \ne \ell_\alpha(x)]$, *and* $\delta(f, \mathcal{LIN}) := \min_\alpha \delta_\alpha$.

In the rest of the section, we prove the BLR soundness using Gowers-Hatami theorem.

Theorem 2.5.2 (Soundness of the BLR test)**.** *For every* $f : \mathbb{F}_q^n \to \mathbb{F}_q$, $\delta(f, \mathcal{LIN}) \le \Pr_{a,b,x,y}[af(x) + bf(y) \ne f(ax + by)]$.

First notice that linearity implies two properties:

1. Additivity $f(x + y) = f(x) + f(y), \quad \forall x, y \in X.$

   Note that for q prime additivity is actually equivalent to linearity. However this is not the case for prime powers $q = p^s$ for $s > 1$.

2. Homogeneity $f(ax) = af(x), \quad \forall x \in X,\ a \in A.$

   For $q = 2$, the two properties are redundant. For $q > 2$, turns out testing homogeneity is crucial for a test to have a high soundness.

In the rest of the section, we discuss an encoding of both properties through a group construction by ChatGPT.

2.5.1 The twisted group G

Define a group G by the set $X \times A$ with the binary operation

$$(x, a) \star (y, b) := (b^{-1}x + a^{-1}y,\ ab). \tag{2.32}$$

Lemma 2.5.3 (G is an abelian group)**.** *The operation $\star$ in (2.32) makes G into an abelian group with identity $e = (0, 1)$ and inverses $(x, a)^{-1} = (-a^2x, a^{-1})$.*

Proof. Associativity. For $(x, a), (y, b), (z, c) \in G$,

$$((x, a) \star (y, b)) \star (z, c) = (b^{-1}x + a^{-1}y,\ ab) \star (z, c) = (b^{-1}c^{-1}x + a^{-1}c^{-1}y + a^{-1}b^{-1}z,\ abc),$$

and an identical computation gives the same expression for $(x, a) \star ((y, b) \star (z, c))$.

Commutativity. $(y, b) \star (x, a) = (a^{-1}y + b^{-1}x,\ ba) = (b^{-1}x + a^{-1}y,\ ab) = (x, a) \star (y, b)$.

Identity. For any $(y, b) \in G$, $(0, 1) \star (y, b) = (b^{-1} \cdot 0 + 1^{-1} \cdot y,\ 1 \cdot b) = (y, b)$.

Inverse. For any $(x, a) \in G$,

$$(x, a) \star (-a^2x, a^{-1}) = ((a^{-1})^{-1} \cdot x + a^{-1} \cdot (-a^2x),\ a \cdot a^{-1}) = (ax - ax,\ 1) = (0, 1).$$

□

It is at first surprising to come up with $\star$ operation, but G is isomorphic to the direct product group.

Proposition 2.5.4 (Group isomorphism to product group)**.** *Let $(G, \star)$ be an abelian group defined by Equation 2.32. The map*

$$\Psi : G \to (X, +) \times (A, \cdot), \qquad \Psi(x, a) := (ax, a), \tag{2.33}$$

is a group isomorphism, where the codomain is the direct product of $(X, +)$ and $(A, \cdot)$ with operation $\star_\Psi : (x,\ a) \star_\Psi (y,\ b) = (x + y,\ ab)$.

Proof. Compute

$$\Psi((x, a) \star (y, b)) = \Psi(b^{-1}x + a^{-1}y,\ ab) = (ab \cdot (b^{-1}x + a^{-1}y),\ ab) = (ax + by,\ ab),$$

which equals $\Psi(x, a) \star_\Psi \Psi(y, b) = (ax + by,\ ab)$. The map Ψ is bijective with inverse $\Psi^{-1}(y, b) = (b^{-1}y, b)$. □

The isomorphism gives us a easy way to compute its irreducible representations which forms the Pontryagin dual group (the set of irrep characters for an abelian group).

Fact 2.5.5 (Pontryagin dual over direct product)**.** *Let G_1, G_2 be finite abelian groups. Let $H = G_1 \times G_2$ be their direct product. Then the Pontryagin dual formed by all the irreps. is $\hat{H} = \hat{G}_1 \times \hat{G}_2$.*

Proposition 2.5.6 (Group characters)**.** *The group $(G, \star)$ has the Pontryagin dual character group*

$$\hat{G} = \{\chi_{\alpha,k}\}_{\alpha \in X, k \in [0,\cdots,q-2]}, \quad \chi_{\alpha,k}(x, a) = \omega^{\langle \alpha, ax \rangle}\psi_k(a), \tag{2.34}$$

where $\psi_k(a) = e^{\frac{i2\pi mk}{q-1}}$ and $a = g_0^m$ for a fixed multiplicative generator $g_0 \in A$.

Proof. By Proposition. 2.5.4, $(G, \star) \cong (X, +) \times (A, \cdot)$ via $\Psi(x, a) = (ax, a)$. First we consider the characters of the product group. Using Fact. 2.5.5, we have

$$\widehat{(X, +) \times (A, \cdot)} = \hat{X} \times \hat{A}.$$

The character groups of X and A are

$$\hat{X} = \{\phi_\alpha(y) = \omega^{\langle \alpha, y \rangle}\}_{\alpha \in X}, \tag{2.35}$$

$$\hat{A} = \{\psi_k(b) = e^{\frac{i2\pi mk}{q-1}}\}_{k \in [0, \cdots, q-2]}, \tag{2.36}$$

where $b = g_0^m$ for a fixed multiplicative generator $g_0 \in A$. Therefore, $\widehat{(X, +) \times (A, \cdot)} = \{\Theta_{\alpha,k}(y, b) = \phi_\alpha(y)\psi_k(b)\}_{\alpha,k}$. The character $\chi \in \hat{G}$ is given by the pull-back through the isomorphism

$$\chi_{\alpha,k}(x, a) = \Theta_{\alpha,k}(\Psi(x, a)) = \Theta_{\alpha,k}(ax, a). \tag{2.37}$$

Therefore, we have $\chi_{\alpha,k}(x, a) = \phi_\alpha(ax)\psi_k(a) = \omega^{\langle \alpha, ax \rangle}\psi_k(a)$. $\square$

2.5.2 Approximate representation and matrix lift of f

Define a unitary matrix valued lift $F_f : G \to U(\mathbb{C}^{q-1})$ by the diagonal matrix

$$F_f(x, a) = D(af(x)) := \sum_{c \in A} \omega^{caf(x)} |c\rangle\langle c|, \tag{2.38}$$

where $\{|c\rangle\}$ denotes the orthonormal basis vectors of a Hilbert space $\mathcal{H} \cong \mathbb{C}^{q-1}$ and the inner product between operators defined by $\langle A, B \rangle_\sigma = \mathrm{Tr}(A^* B \sigma)$ and choose $\sigma = \mathbb{I}/(q-1)$. We first show that the BLR condition indeed gives an approximate represenatation.

Lemma 2.5.7 (Lift operation from BLR condition)**.** *Let $f : X \to \mathbb{F}_q$ and $(x, a), (y, b) \in G$ be rejected by the BLR test with probability ε, then the unitary matrix lift F_f in Equation 2.38 is an $(\frac{q}{q-1}\varepsilon, \frac{\mathbb{I}}{q-1})$ approximate representation of $(G, \star)$.*

Proof. The BLR test condition gives

$$\Pr_{c,d,x,y}\big[cf(x) + df(y) \neq f(cx + dy)\big] \leq \varepsilon.$$

First we observe that

$$\Pr_{g,h \in G}\big[F_f(g \star h) = F_f(g)F_f(h)\big] = \Pr_{c,d,x,y}\big[cf(x) + df(y) = f(cx + dy)\big]. \tag{2.39}$$

Using Equation 2.32 and Equation 2.38,

$$F_f\big((x, a) \star (y, b)\big) = \sum_{c \in A} \omega^{cab\, f(b^{-1}x + a^{-1}y)} |c\rangle\langle c|,$$

$$F_f(x, a)\, F_f(y, b) = \sum_{c \in A} \omega^{c(af(x) + bf(y))} |c\rangle\langle c|.$$

These coincide iff their exponents agree modulo p Check this is true if we add Trace:

$$ab\, f(b^{-1}x + a^{-1}y) = af(x) + bf(y).$$

Since a, b have a multiplicative inverse, substitute $c := b^{-1}$, $d := a^{-1}$, we have

$$f(cx+dy) = cf(x) + df(y),$$

which proves Equation 2.39.

To relate the test condition to the approximate representation, first note the equivalent definition of approximate representation

$$\mathbb{E}_{g,h\in G}\Re\left\langle F_f(g)^* F_f(h), F_f\left(g^{-1}h\right)\right\rangle_\sigma = \mathbb{E}_{g,h\in G}\Re\left\langle F_f(g) F_f(h), F_f\left(gh\right)\right\rangle_\sigma, \tag{2.40}$$

because F_f is unitary so $F_f(g)^* = F_f(g^{-1}), \forall g \in G$. Then evaluate the right hand side

$$\begin{aligned}
\mathbb{E}_{g,h\in G}\Re\left\langle F_f(g) F_f(h), F_f\left(gh\right)\right\rangle_\sigma &= \mathbb{E}_{g,h\in G}\Re\mathrm{Tr}\left[F_f^*(h) F_f^*(g) F_f\left(gh\right)\sigma\right], \\
&= \mathbb{E}_{(x,a),(y,b)\in G}\frac{1}{q-1}\Re\mathrm{Tr}\left[\sum_{c\in A}\omega^{-c(af(x)+bf(y))}\omega^{cab\,f(b^{-1}x+a^{-1}y)}|c\rangle\langle c|\right], \\
&= \mathbb{E}_{(x,a),(y,b)\in G}\frac{1}{q-1}\Re\sum_{c\in A}\omega^{cz}, \qquad (2.41)
\end{aligned}$$

where we define $z = c(ab\,f(b^{-1}x + a^{-1}y) - (af(x) + bf(y)))$. To compute the expectation, we can split to two cases

$$\sum_{c\in A}\omega^{cz} = \begin{cases} q-1, & z = 0 \quad \text{when } f \text{ passes BLR,} \\ -1, & z \neq 0 \quad \text{when } f \text{ fails BLR.}\end{cases} \tag{2.42}$$

The second case is because $\sum_{c\in A}\omega^{cz} + 1 = \sum_{c\in\mathbb{F}_q}\omega^{cz} = 0$. Therefore, we have the approximate representation

$$\mathbb{E}_{g,h\in G}\Re\left\langle F_f(g) F_f(h), F_f\left(gh\right)\right\rangle_\sigma \geq (1-\varepsilon) - \frac{1}{q-1}\varepsilon = 1 - \frac{q}{q-1}\varepsilon. \tag{2.43}$$

Hence F_f is an approximate representation with $\epsilon = \frac{q}{q-1}\varepsilon$ as claimed. ◻

The approximate representation "appears" slightly worse than what we would have hoped $\epsilon > \varepsilon$, but we will see this is not an issue for achieving a high soundness. Remarkably, the twisted group operation let us recover the linearity test condition exactly.

Fact 2.5.8. *F_f is an exact representation of $(G, \star)$ if and only if $f \in \mathcal{LIN}$.*

Proof. If F_f is an exact representation of G, then Equation 2.39 forces $\varepsilon = 0$. Setting $c = d = 1$ in the BLR condition gives $f(x+y) = f(x) + f(y)$ for all x, y, so f is additive. Set $y = 0$ and $c = 1 \in A$: $f(x) = f(x) + d\,f(0)$ for any $d \in A$. Set $y = 0$ and $d = 1$, then $f(0) = 0$ and $f(cx) = cf(x)$. Combined with additivity, f is $\mathbb{F}_q$-linear, so $f \in \mathcal{LIN}$.

Conversely, if $f = \ell_\alpha$ for some $\alpha \in X$, then so F_f is a group homomorphism and an exact representation of G. ◻

2.5.3 Gowers-Hatami implies BLR high soundness

In this section, we first prove the key lemma relating the implication of GH theorem to the hamming distance. We provide two proofs:

1. The first proof in Lemma 2.5.9 should only use the existence of an isometry V and exact representation R. Unfortunately, we ended up with an extra constraint about the range of ϵ.

2. In the second proof in Lemma 2.5.11, we use an explicit isometry V constructed via F_f in the proof of Gowers-Hatami theorem. This proof might require a extra care in formalization of lean. However, this hybrid Fourier-GH approach gives us an unconditional high soundness of the linear BLR test.

Finally, we provide a proof of Theorem 2.5.1.

Lemma 2.5.9 (Gowers-Hatami to linearity (Convex character proof))**.** *Suppose $F_f : G \to U(\mathbb{C}^{q-1})$ defined in Equation 2.38 via $f : \mathbb{F}_q^n \to \mathbb{F}_q$ is an (ϵ, σ) approximate unitary representation of $(G, \star)$ defined in Equation 2.32. Then, there exists a linear function $\ell : \mathbb{F}_q^n \to \mathbb{F}_q$ such that the relative Hamming distance $\delta(f, \ell)$ is bounded by:*

$$\delta(f, \ell) \leq \frac{q-1}{q}\epsilon. \tag{2.44}$$

Proof. By Gowers-Hatami theorem, we have an isometry $V : \mathcal{H} \to \mathcal{H}'$ and an exact representation $R : G \to \mathrm{End}(\mathcal{H}')$, where the Hilbert space $\mathcal{H} \cong \mathbb{C}^{q-1}$ and the space R acts on $\mathcal{H}' = L(G, \mathcal{H})$

$$\mathbb{E}_{g\in G}\mathfrak{R}\langle F_f(g), V^*R(g)V\rangle_\sigma \geq 1 - \epsilon. \tag{2.45}$$

Let $\{|c\rangle\}_{c\in\mathbb{F}_q^\times}$ be the orthonormal basis for $\mathcal{H}$, and the set of characters $\{\chi\}_{\chi\in\hat{G}}$ be an orthonormal basis for $L(G, \mathbb{C})$.

Because $F_f(g)$ is diagonal in $\{|c\rangle\}$ with entries $\omega^{caf(x)}$, we can rewrite the inner product trace:

$$\langle F_f(g), V^*R(g)V\rangle_\sigma = \frac{1}{q-1}\sum_{c\in\mathbb{F}_q^\times}\langle c|F_f^*(g)V^*R(g)V|c\rangle = \frac{1}{q-1}\sum_{c\in\mathbb{F}_q^\times}\omega^{-caf(x)}\langle v_c, R(g)v_c\rangle_{L(G,\mathcal{H})},$$

where $|v_c\rangle = V|c\rangle \in \mathcal{H}'$. Taking the expectation over $g = (x, a)$ and taking the real part, condition Equation 2.64 becomes:

$$\frac{1}{q-1}\sum_{c\in\mathbb{F}_q^\times}\mathbb{E}_{x,a}\mathfrak{R}\left[\omega^{-caf(x)}\langle v_c, R(x,a)v_c\rangle\right] \geq 1 - \epsilon. \tag{2.46}$$

A basis for $L(G, \mathcal{H})$ is given by $\{\chi|c\rangle\}_{\chi\in\hat{G},c\in\mathbb{F}_q^\times}$. By Proposition. 2.5.6, the characters are given by

$$\hat{G} = \{\chi_{\alpha,k}\}_{\alpha\in X,k\in[0,\cdots,q-2]}, \chi_{\alpha,k}(x,a) = \omega^{\langle\alpha,ax\rangle}\psi_k(a).$$

The regular representation $R(g)$ acts on $L(G, \mathcal{H})$ so it can be decomposed as

$$R(g) = \sum_{\alpha,k}\chi_{\alpha,k}(g)P_{\alpha,k}, \tag{2.47}$$

where $P_{\alpha,k} = |\chi_{\alpha,k}\rangle\langle\chi_{\alpha,k}| \otimes \mathbb{I}_{\mathcal{H}}$ projects onto the character function subspace (this leaves $\mathcal{H}$ invariant).

Substituting this into the matrix element for v_c, we obtain:

$$\langle v_c, R(g)v_c\rangle = \sum_{\alpha,k}\chi_{\alpha,k}(g)\langle v_c, P_{\alpha,k}v_c\rangle.$$

Let $p_{c,\alpha,k} := \langle v_c, P_{\alpha,k} v_c \rangle$. As $P_{\alpha,k}$ is a projector, then

$$p_{c,\alpha,k} = \langle v_c | \chi_{\alpha,k} \rangle \langle \chi_{\alpha,k} | \otimes \mathbb{I}_{\mathcal{H}} | v_c \rangle \geq 0. \tag{2.48}$$

Since V is an isometry $V^* V = \mathbb{I}_{\mathcal{H}}$, we have $\langle c | V^* V | c \rangle = \langle c, c \rangle_{\mathcal{H}} = 1$. Equivalently, we have $\langle v_c, v_c \rangle_{L(G,\mathcal{H})} = 1$. Since $\sum P_{\alpha,k} = I_{L(G,H)}$, it follows that $p_{c,\alpha,k}$ is a valid probability distribution over the characters,

$$\sum_{\alpha,k} p_{c,\alpha,k} = 1, \quad \forall c \in \mathbb{F}_q^\times. \tag{2.49}$$

Substituting the spectral decomposition Equation 2.67 back into Equation 2.65, we get:

$$\sum_{c \in \mathbb{F}_q^\times} \left(\frac{1}{q-1} \right) \sum_{\alpha,k} p_{c,\alpha,k} \Re \left(\mathbb{E}_{x,a} \left[\omega^{-caf(x)} \chi_{\alpha,k}(x,a) \right] \right) \geq 1 - \epsilon.$$

This is a convex combination (a weighted average) of real values less than 1. Because the average is at least $1-\epsilon$, there must exist at least one specific configuration (c^*, α^*, k^*) with $c^* \neq 0$ such that its individual value satisfies:

$$\Re \left(\mathbb{E}_{x,a} \left[\omega^{-c^* a f(x)} \omega^{\langle \alpha^*, ax \rangle} \psi_{k^*}(a) \right] \right) \geq 1 - \epsilon. \tag{2.50}$$

Because $c^* \in \mathbb{F}_q^\times$ is invertible, we define a specific linear function $\ell(x) := \langle (c^*)^{-1} \alpha^*, x \rangle$. Notice that $\langle \alpha^*, ax \rangle = c^* a \ell(x)$. We can rewrite the argument of the expectation in Equation 2.74 as:

$$E^* := \mathbb{E}_x \left[\mathbb{E}_a \left[\omega^{c^* a(\ell(x) - f(x))} \psi_{k^*}(a) \right] \right].$$

For the rest of this proof, without using specific form of $p_{c,\alpha,k}$, we consider two cases: $k^* = 0$ and $k^* \neq 0$. Define $z(x) = c^*(\ell(x) - f(x))$.

1. Consider $k^* = 0$ so that $\psi_0(a) = 1$, the inner expectation over $a \in \mathbb{F}_q^\times$ for a fixed x:

$$\mathbb{E}_{a \in \in \mathbb{F}_q^\times} \omega^{c^* a(\ell(x) - f(x))} = \begin{cases} 1, & z(x) = 0, \\ -\frac{1}{q-1}, & z(x) \neq 0, \end{cases} \tag{2.51}$$

where in the second case we have again used average of roots of unity is $\sum_{b \in \in \mathbb{F}_q^\times} \omega^b + 1 = 0$.

Evaluating the outer expectation over x and recall the definition of hamming distance

$$\delta_\alpha := \Pr_x [f(x) \neq \ell_\alpha(x)], \tag{2.52}$$

we find:

$$\Re E^* = E^* = 1 \cdot (1 - \delta(f,\ell)) + \left(\frac{-1}{q-1} \right) \cdot \delta(f,\ell) = 1 - \delta(f,\ell) \left(1 + \frac{1}{q-1} \right) = 1 - \delta(f,\ell) \left(\frac{q}{q-1} \right).$$

Applying our bound from Equation 2.74, we have:

$$1 - \delta(f,\ell) \left(\frac{q}{q-1} \right) \geq 1 - \epsilon.$$

Rearranging yields the final result:

$$\delta(f,\ell) \leq \frac{q-1}{q} \epsilon.$$

2. Consider $k^* \neq 0$, then by Proposition. 2.5.6 we have $\psi_k(a) = e^{\frac{i2\pi mk}{q-1}}$ and $a = g_0^m$ for a fixed multiplicative generator $g_0 \in \mathbb{F}_q^\times$. Note that averaging over $\psi_k(a)$ we have

$$\mathbb{E}_{a\in\mathbb{F}_q^\times}\psi_k(a) = \frac{1}{q-1}\sum_{m=0}^{q-2} e^{\frac{i2\pi mk}{q-1}} = 0. \tag{2.53}$$

The average expectation over $a \in \mathbb{F}_q^\times$ for a fixed x gives:

$$\mathbb{E}_{a\in\in\mathbb{F}_q^\times}\omega^{c^*a(\ell(x)-f(x))}\psi_{k^*}(a) = \begin{cases} 0, & z(x) = 0, \\ \frac{1}{q-1}\sum_{a\in\in\mathbb{F}_q^\times}\omega^{az(x)}\psi_{k^*}(a), & z(x) \neq 0. \end{cases} \tag{2.54}$$

In the second case, let $b = az(x)$ so that

$$\begin{aligned} \frac{1}{q-1}\sum_{a\in\in\mathbb{F}_q^\times}\omega^{az(x)}\psi_{k^*}(a) &= \frac{1}{q-1}\sum_{b\in\in\mathbb{F}_q^\times}\omega^b\psi_{k^*}(z^{-1}b), \\ &= \frac{\psi_{k^*}(z^{-1})}{q-1}\sum_{b\in\in\mathbb{F}_q^\times}\omega^b\psi_{k^*}(b). \end{aligned} \tag{2.55}$$

It turns out the summation is a known quantitiy called *the Gauss sum*

$$\mathcal{G}(\psi_{k^*}) = \sum_{b\in\in\mathbb{F}_q^\times}\omega^b\psi_{k^*}(b). \tag{2.56}$$

Is it believed that there is no analytic expression for $\mathcal{G}$ but it is known that

$$|\mathcal{G}(\psi_{k^*})| = \sqrt{q} \tag{2.57}$$

Evaluating the outer expectation over x and recall the definition of hamming distance gives

$$\Re E^* = 0\cdot(1-\delta(f,\ell)) + \Re\left(\frac{\psi_{k^*}(z^{-1})}{q-1}\mathcal{G}(\psi_{k^*})\right)\cdot\delta(f,\ell).$$

Therefore we have

$$\Re E^* = \Re\left(\frac{\psi_{k^*}(z^{-1})}{q-1}\mathcal{G}(\psi_{k^*})\right)\delta(f,\ell) \leq \frac{|\mathcal{G}(\psi_{k^*})|}{q-1} = \frac{\sqrt{q}}{q-1}.$$

On the other hand, from Equation 2.74 we have

$$\Re E^* \geq 1-\epsilon.$$

This is a contradiction to the existence of k^* if

$$\frac{\sqrt{q}}{q-1} < 1-\epsilon. \tag{2.58}$$

Then if we have a good test outcome such that Equation 2.58 is satisfied, then we only have a contribution from $k^* = 0$, where we have already achieved the bound.

□

The above bound is tight assuming we pass the test with high probability such that $\epsilon < 1-\frac{\sqrt{q}}{q-1}$. Next, we try to give a proof that is unconditional, using Fourier coefficients.

First let us write down statements about Fourier coefficients of F_f, which we will use in the proof.

Proposition 2.5.10 (Fourier coefficient of F_f)**.** *Let the matrix lift $F_f : G \to U(\mathbb{C}^{q-1})$ be defined by the diagonal matrix*

$$F_f(x, a) = D(af(x)) := \sum_{c\in A} \omega^{caf(x)}|c\rangle\langle c|,$$

then its Fourier coefficients can be written as

$$[\widehat{F}_f(\chi_{c\beta,k})]_{cc} = \psi_k(c)\nu_k(\beta), \tag{2.59}$$

where

$$\nu_k(\beta) := \mathbb{E}_{x,b}\left[\omega^{b(f(x)-\langle\beta,x\rangle)}\overline{\psi_k(b)}\right], \tag{2.60}$$

Proof. By Proposition. 2.5.6, the characters are given by

$$\widehat{G} = \{\chi_{\alpha,k}\}_{\alpha\in X, k\in[0,\cdots,q-2]}, \chi_{\alpha,k}(x, a) = \omega^{\langle\alpha,ax\rangle}\psi_k(a).$$

Therefore the Fourier coefficients of F_f are

$$\begin{aligned}\widehat{F}_f(\chi_{\alpha,k}) &= \sum_{c\in\mathbb{F}_q^\times} \mathbb{E}_{x,a}\omega^{caf(x)}\bar{\chi}_{\alpha,k}(x, a),\\ &= \sum_{c\in\mathbb{F}_q^\times} \mathbb{E}_{x,a}\left[\omega^{a(cf(x)-\langle\alpha,x\rangle)}\overline{\psi_k(a)}\right]|c\rangle\langle c|. \end{aligned}\tag{2.61}$$

Since $c \in \mathbb{F}_q^\times$ is invertible, we perform a change of variables. Let $\alpha = c\beta$ for a fresh $\beta \in \mathbb{F}_q^n$. Because c is fixed, this is a bijection on $\mathbb{F}_q^n$. We also substitute $b = ca$, which is a bijection on $\mathbb{F}_q^\times$. This means $a = bc^{-1}$, and by the multiplicity of Dirichlet characters, $\overline{\psi_k(a)} = \overline{\psi_k(bc^{-1})} = \overline{\psi_k(b)}\psi_k(c)$.

Substituting these into the expectation isolates the c-dependence:

$$\begin{aligned}[\widehat{F}_f(\chi_{c\beta,k})]_{cc} &= \mathbb{E}_{x,b}\left[\omega^{b(f(x)-\langle\beta,x\rangle)}\overline{\psi_k(b)}\psi_k(c)\right]\\ &= \psi_k(c)\nu_k(\beta), \end{aligned}\tag{2.62}$$

where $\nu_k(\beta) := \mathbb{E}_{x,b}\left[\omega^{b(f(x)-\langle\beta,x\rangle)}\overline{\psi_k(b)}\right]$ is a c-independent coefficient. □

Lemma 2.5.11 (Gowers-Hatami to linearity (Convex character Fourier hybrid))**.** *Suppose $F_f : G \to U(\mathbb{C}^{q-1})$ defined in Equation 2.38 via $f : \mathbb{F}_q^n \to \mathbb{F}_q$ is an (ϵ, σ) approximate unitary representation of $(G, \star)$ defined in Equation 2.32. Then, there exists a linear function $\ell : \mathbb{F}_q^n \to \mathbb{F}_q$ such that the relative Hamming distance $\delta(f, \ell)$ is bounded by:*

$$\delta(f, \ell) \le \frac{q-1}{q}\epsilon. \tag{2.63}$$

Proof. By Gowers-Hatami theorem, we have an isometry $V : \mathcal{H} \to \mathcal{H}'$ and an exact representation $R : G \to \mathrm{End}(\mathcal{H}')$, where the Hilbert space $\mathcal{H} \cong \mathbb{C}^{q-1}$ and the space R acts on $\mathcal{H}' = L(G, \mathcal{H})$

$$\mathbb{E}_{g\in G}\Re\langle F_f(g), V^*R(g)V\rangle_\sigma \ge 1-\epsilon. \tag{2.64}$$

Let $\{|c\rangle\}_{c\in\mathbb{F}_q^\times}$ be the orthonormal basis for $\mathcal{H}$, and the set of characters $\{\chi\}_{\chi\in\hat{G}}$ be an orthonormal basis for $L(G,\mathbb{C})$.

Because $F_f(g)$ is diagonal in $\{|c\rangle\}$ with entries $\omega^{caf(x)}$, we can rewrite the inner product trace:

$$\langle F_f(g), V^*R(g)V\rangle_\sigma = \frac{1}{q-1}\sum_{c\in\mathbb{F}_q^\times}\langle c|F_f^*(g)V^*R(g)V|c\rangle = \frac{1}{q-1}\sum_{c\in\mathbb{F}_q^\times}\omega^{-caf(x)}\langle v_c, R(g)v_c\rangle_{L(G,\mathcal{H})},$$

where $|v_c\rangle = V|c\rangle \in \mathcal{H}'$. Taking the expectation over $g=(x,a)$ and taking the real part, condition Equation 2.64 becomes:

$$\frac{1}{q-1}\sum_{c\in\mathbb{F}_q^\times}\mathbb{E}_{x,a}\Re\left[\omega^{-caf(x)}\langle v_c, R(x,a)v_c\rangle\right] \geq 1-\epsilon. \tag{2.65}$$

A basis for $L(G,\mathcal{H})$ is given by $\{\chi|c\rangle\}_{\chi\in\hat{G},c\in\mathbb{F}_q^\times}$. By Proposition. 2.5.6, the characters are given by

$$\hat{G} = \{\chi_{\alpha,k}\}_{\alpha\in X,k\in[0,\cdots,q-2]}, \chi_{\alpha,k}(x,a) = \omega^{\langle\alpha,ax\rangle}\psi_k(a).$$

Moreover, the Fourier coefficients of F_f are

$$\begin{aligned}\hat{F}_f(\chi_{\alpha,k}) &= \sum_{c\in\mathbb{F}_q^\times}\mathbb{E}_{x,a}\omega^{caf(x)}\bar{\chi}_{\alpha,k}(x,a),\\ &= \sum_{c\in\mathbb{F}_q^\times}\mathbb{E}_{x,a}\left[\omega^{a(cf(x)-\langle\alpha,x\rangle)}\overline{\psi_k(a)}\right]|c\rangle\langle c|.\end{aligned} \tag{2.66}$$

The regular representation $R(g)$ acts on $L(G,\mathcal{H})$ so it can be decomposed as

$$R(g) = \sum_{\alpha,k}\chi_{\alpha,k}(g)P_{\alpha,k}, \tag{2.67}$$

where $P_{\alpha,k} = |\chi_{\alpha,k}\rangle\langle\chi_{\alpha,k}|\otimes\mathbb{I}_{\mathcal{H}}$ projects onto the character function subspace (this leaves $\mathcal{H}$ invariant).

Substituting this into the matrix element for v_c, we obtain:

$$\langle v_c, R(g)v_c\rangle = \sum_{\alpha,k}\chi_{\alpha,k}(g)\langle v_c, P_{\alpha,k}v_c\rangle.$$

Let $p_{c,\alpha,k} := \langle v_c, P_{\alpha,k}v_c\rangle$. As $P_{\alpha,k}$ is a projector, then

$$p_{c,\alpha,k} = \langle v_c|\chi_{\alpha,k}\rangle\langle\chi_{\alpha,k}|\otimes\mathbb{I}_{\mathcal{H}}|v_c\rangle \geq 0. \tag{2.68}$$

Let the isometry be given by $V|c\rangle = [F_f(g)]_c$, then the matrix element is given by the Fourier coefficients

$$p_{c,\alpha,k} = \left|[\hat{F}_f(\chi_{\alpha,k})]_{cc}\right|^2. \tag{2.69}$$

By Proposition. 2.5.10, let $\alpha = c\beta$, we can show

$$p_{c,c\beta,k} = |\psi_k(c)\nu_k(\beta)|^2 = |\nu_k(\beta)|^2, \tag{2.70}$$

where

$$\nu_k(\beta) := \mathbb{E}_{x,b}\left[\omega^{b(f(x)-\langle\beta,x\rangle)}\overline{\psi_k(b)}\right],$$

is defined in Equation 2.60.

Since V is an isometry $V^*V = \mathbb{I}_{\mathcal{H}}$, we have $\langle c|V^*V|c\rangle = \langle c, c\rangle_{\mathcal{H}} = 1$. Equivalently, we have $\langle v_c, v_c\rangle_{L(G,\mathcal{H})} = 1$. Since $\sum P_{\alpha,k} = I_{L(G,H)}$, it follows that $p_{c,\alpha,k}$ is a valid probability distribution over the characters,

$$\sum_{\alpha,k} p_{c,\alpha,k} = 1, \quad \forall c \in \mathbb{F}_q^\times. \tag{2.71}$$

Equivalently, we have the relabeled probability

$$\sum_{\beta,k} p_{c,c\beta,k} = 1, \quad \forall c \in \mathbb{F}_q^\times. \tag{2.72}$$

Therefore $|\nu_k(\beta)|^2$ is a probability distribution indexed by (k, β). Because the $k = 0$ probability is a subset, we have

$$\sum_{\beta\in\mathbb{F}_q^n} |\nu_0(\beta)|^2 \leq \sum_{\beta\in\mathbb{F}_q^n} \sum_k |\nu_k(\beta)|^2 = 1. \tag{2.73}$$

Substituting the spectral decomposition Equation 2.67 back into Equation 2.65, we define:

$$\begin{aligned} E &:= \frac{1}{q-1} \sum_{c\in\mathbb{F}_q^\times} \mathbb{E}_{x,a} \Re \left[\omega^{-caf(x)} \langle v_c, R(x,a) v_c\rangle\right], \\ &= \frac{1}{q-1} \sum_{c\in\mathbb{F}_q^\times} \sum_{\alpha,k} p_{c,\alpha,k} \Re \left(\mathbb{E}_{x,a}\left[\omega^{-caf(x)} \omega^{\langle \alpha, ax\rangle} \psi_k(a)\right]\right). \end{aligned}$$

Relabel the sum by $\alpha = c\beta$ and $b = ca$, we have

$$E = \frac{1}{q-1} \sum_{c\in\mathbb{F}_q^\times} \sum_{\beta,k} |\nu_k(\beta)|^2 \Re \left(\mathbb{E}_{x,b}\left[\omega^{b(\langle\beta,x\rangle - f(x))} \psi_k(c^{-1}) \psi_k(b)\right]\right).$$

Use the indicator function of $\mathbb{F}_q^\times$:

$$\sum_{c\in\mathbb{F}_q^\times} \psi_k(c^{-1}) = (q-1)\mathbb{I}(k=0)$$

we have

$$E = \sum_{\beta} |\nu_0(\beta)|^2 \Re \left(\mathbb{E}_{x,b}\left[\omega^{b(\langle\beta,x\rangle - f(x))}\right]\right).$$

This is a convex combination (a weighted average) of real values less than 1 with the weight sum to less than 1 by Equation 2.73. Because the average is at least $E \geq 1-\epsilon$, there must exist at least one specific configuration β^* such that its individual value satisfies:

$$E^* := \Re \left(\mathbb{E}_{x,b}\left[\omega^{b(\langle\beta^*,x\rangle - f(x))}\right]\right) \geq 1-\epsilon. \tag{2.74}$$

Let $\ell_\beta(x) := \langle \beta^*, x\rangle$, the inner expectation over $b \in \mathbb{F}_q^\times$ for a fixed x:

$$\mathbb{E}_{b\in\in\mathbb{F}_q^\times} \omega^{bz(x)} = \begin{cases} 1, & z(x) = 0, \\ -\frac{1}{q-1}, & z(x) \neq 0, \end{cases} \tag{2.75}$$

where in the second case we have again used average of roots of unity is $\sum_{b\in\in\mathbb{F}_q^\times} \omega^b + 1 = 0$.

Evaluating the outer expectation over x and recall the definition of hamming distance

$$\delta(f, \ell_\alpha) := \Pr_x[f(x) \neq \ell_\alpha(x)], \tag{2.76}$$

we find:

$$E^* = 1 \cdot (1 - \delta(f, \ell_\beta)) + \left(\frac{-1}{q-1}\right) \cdot \delta(f, \ell_\beta) = 1 - \delta(f, \ell_\beta)\left(1 + \frac{1}{q-1}\right) = 1 - \delta(f, \ell_\beta)\left(\frac{q}{q-1}\right).$$

Applying our bound from Equation 2.74, we have:

$$1 - \delta(f, \ell_\beta)\left(\frac{q}{q-1}\right) \geq 1 - \epsilon.$$

Rearranging yields the final result: $\exists \ell_\beta = \langle \beta^*, x \rangle$ such that

$$\delta(f, \ell_\beta) \leq \frac{q-1}{q}\epsilon.$$

□

Finally, we are ready to prove the main theorem of BLR linearity test soundness.

Theorem 2.5.1 (Soundness of the BLR test)**.** *Let q be prime. For every $f : \mathbb{F}_q^n \to \mathbb{F}_q$, $\delta(f, \mathcal{LIN}) \leq \Pr_{a,b,x,y}[af(x) + bf(y) \neq f(ax + by)]$.*

Proof. First denote

$$X := \mathbb{F}_q^n, \qquad A := \mathbb{F}_q^\times, \qquad \omega := e^{2\pi i/p}.$$

Define a group G by the set $X \times A$ with the binary operation

$$(x, a) \star (y, b) := (b^{-1}x + a^{-1}y,\ ab). \tag{2.77}$$

Let F_f be a unitary matrix valued lift $F_f : G \to U(\mathbb{C}^{q-1})$ by the diagonal matrix

$$F_f(x, a) = D(af(x)) := \sum_{c \in A} \omega^{caf(x)} |c\rangle\langle c|, \tag{2.78}$$

where $\{|c\rangle\}$ denotes the orthonormal basis vectors of a Hilbert space $\mathcal{H} \cong \mathbb{C}^{q-1}$ and the inner product between operators defined by $\langle A, B\rangle_\sigma = \mathrm{Tr}(A^* B\sigma)$ and choose $\sigma = \mathbb{I}/(q-1)$. By Lemma 2.5.7, since f is rejected with probability ε, then F_f is an $(\frac{q}{q-1}\varepsilon, \frac{\mathbb{I}}{q-1})$ approximate representation of $(G, \star)$.

Then by Lemma 2.5.11, there exists a linear function $\ell : \mathbb{F}_q^n \to \mathbb{F}_q$ such that the relative Hamming distance $\delta(f, \ell)$ is bounded by:

$$\delta(f, \ell) \leq \frac{q-1}{q}\frac{q}{q-1}\varepsilon = \varepsilon. \tag{2.79}$$

This proves the claim

$$\delta(f, \mathcal{LIN}) := \min_\alpha \delta(f, \ell_\alpha) \leq \delta(f, \ell) \leq \varepsilon.$$

$\square$

Miraculously, the loose factor in $(\epsilon' = \frac{q}{q-1}\varepsilon, \frac{\mathbb{I}}{q-1})$ approximate representation F_f cancels with the tighter factor in

$$\delta(f, \ell) \leq \frac{q-1}{q}\epsilon'.$$

Chapter 3

Exponential-length PCP

Definition 3.0.1. *A system of m quadratic equations in n variables over a field $\mathbb{F}$ is a list of polynomials $p_1, \dots, p_m \in \mathbb{F}[x_1, \dots, x_n]$ where each p_i has total degree at most 2.*

$$\mathrm{QESAT}(\mathbb{F}, n) := \{(p_1, \dots, p_m) \mid \exists a_1, \dots, a_n \in \mathbb{F},\ \forall i \in [m],\ p_i(a_1, \dots, a_n) = 0\}.$$

For example, $(x+1, xy+z) \in \mathrm{QESAT}(\mathbb{F}_2, 3)$.

Definition 3.0.2. *A* PCP verifier *is a probabilistic oracle Turing machine V with query access to a function $\pi : [\ell(|x|)] \to \Sigma$.*

Definition 3.0.3. *A language $L \subseteq \{0,1\}^*$ is in* $\mathrm{PCP}[\varepsilon_c, \varepsilon_s, \Sigma, \ell, q, r]$ *if there exists a polynomial-time PCP verifier V such that for every $x \in \{0,1\}^*$, V makes at most $q(|x|)$ queries to the proof oracle, uses at most $r(|x|)$ bits of randomness, and the following holds:*

- *Completeness: If $x \in L$, then $\exists \pi \in \Sigma^{\ell(|x|)}$, $\Pr\left[V^{\pi}(x) = 1\right] \geq 1 - \varepsilon_c$.*
- *Soundness: If $x \notin L$, then $\forall \tilde{\pi} \in \Sigma^{\ell(|x|)}$, $\Pr\left[V^{\tilde{\pi}}(x) = 1\right] \leq \varepsilon_s$.*

The following theorem is the main goal of this chapter.

Theorem 3.0.4.

$$\mathrm{QESAT}(\mathbb{F}_2, n) \in \mathrm{PCP}[\varepsilon_c = 0, \varepsilon_s = 1/2, \Sigma = \mathbb{F}_2, \ell = 2^{|x|}, q = O(1), r = |x|^{O(1)}].$$

Proof. TODO ☐

Definition 3.0.5. *A* LPCP verifier *is a probabilistic oracle Turing machine V with query access to a linear function $\langle \pi, \cdot \rangle : \Sigma^{\ell(|x|)} \to \Sigma$.*

Definition 3.0.6. *A language $L \subseteq \{0,1\}^*$ is in* $\mathrm{LPCP}[\varepsilon_c, \varepsilon_s, \Sigma, \ell, q, r]$ *if there exists a polynomial-time LPCP verifier V such that for every $x \in \{0,1\}^*$, V makes at most $q(|x|)$ queries to the linear proof oracle, uses at most $r(|x|)$ bits of randomness, and the following holds:*

- *Completeness: If $x \in L$, then $\exists \pi \in \Sigma^{\ell(|x|)}$, $\Pr\left[V^{\langle \pi, \cdot \rangle}(x) = 1\right] \geq 1 - \varepsilon_c$.*
- *Soundness: If $x \notin L$, then $\forall \tilde{\pi} \in \Sigma^{\ell(|x|)}$, $\Pr\left[V^{\langle \tilde{\pi}, \cdot \rangle}(x) = 1\right] \leq \varepsilon_s$.*

3.1 Simpler BLR verifier for $\mathbb{F}_2$

Definition 3.1.1. *Let $V^f_{\mathrm{BLR}(\mathbb{F}_2,n)}$ be the verifier with oracle access to $f : \mathbb{F}_2^n \to \mathbb{F}_2$ defined as follows:*

1. *Sample $x, y \leftarrow \mathbb{F}^n$.*
2. *Accept if and only if $f(x) + f(y) = f(x + y)$.*

Theorem 3.1.2. *Let $n > 0$. For every linear function $f : \mathbb{F}_2^n \to \mathbb{F}_2$, we have*

$$\Pr\left[V^f_{\mathrm{BLR}(\mathbb{F}_2,n)} = 1\right] = 1.$$

Proof. □

Theorem 3.1.3. *Let $n > 0$ and $f : \mathbb{F}_2^n \to \mathbb{F}_2$.*

$$\Pr\left[V^f_{\mathrm{BLR}(\mathbb{F}_2,n)} = 0\right] \geq \Delta(f, \mathcal{LIN}).$$

Proof. □

3.2 Linear PCP for linear equations

Definition 3.2.1. *For a field $\mathbb{F}$ and parameters $m, n \in \mathbb{N}$, we define the language*

$$\mathrm{LINEQ}(\mathbb{F}, m, n) := \{(M, c) \in \mathbb{F}^{m\times n} \times \mathbb{F}^m \mid \exists b \in \mathbb{F}^n,\, Mb = c\}.$$

Definition 3.2.2. *Let $V^{\langle b,\cdot\rangle}_{\mathrm{LINEQ}(\mathbb{F},m,n)}(M, c)$ be the LPCP verifier defined as follows:*

1. *Sample $r \leftarrow \mathbb{F}^m$.*
2. *Query $b \in \mathbb{F}^n$ at $u := M^\top r \in \mathbb{F}^n$.*
3. *Accept if and only if $\langle b, u\rangle = \langle c, r\rangle$.*

Theorem 3.2.3.

$$\mathrm{LINEQ}(\mathbb{F}, m, n) \in \mathrm{LPCP}\left[\varepsilon_c = 0, \varepsilon_s = 1/|\mathbb{F}|, \Sigma = \mathbb{F}, \ell = |x|, q = 1, r = m \log |\mathbb{F}|\right].$$

Proof. Consider the LPCP verifier $V^{\langle b,\cdot\rangle}_{\mathrm{LINEQ}(\mathbb{F},m,n)}(M, c)$.
Completeness: If $Mb = c$, then for every $r \in \mathbb{F}^m$, we have

$$\langle b, u\rangle = b^\top(M^\top r) = (Mb)^\top r = \langle c, r\rangle.$$

Soundness: If $Mb \neq c$, then

$$\begin{aligned}\Pr_{r\leftarrow\mathbb{F}^m}\left[\langle b, u\rangle = \langle c, r\rangle\right] &= \Pr_{r\leftarrow\mathbb{F}^m}\left[\langle Mb, r\rangle = \langle c, r\rangle\right]\\ &= \Pr_{r\leftarrow\mathbb{F}^m}\left[\sum_{i=1}^m (Mb - c)_i \cdot r_i = 0\right] \leq 1/|\mathbb{F}|,\end{aligned}$$

where the last inequality follows from the Schwartz-Zippel lemma. □

3.3 Linear PCP for tensor checks

Definition 3.3.1. *For a field $\mathbb{F}$, vectors $a \in \mathbb{F}^n$ and $b \in \mathbb{F}^m$, the tensor product $a \otimes b \in \mathbb{F}^{n \times m}$ is the matrix defined by*

$$(a \otimes b)_{i,j} := a_i \cdot b_j \quad \forall i \in [n],\, j \in [m].$$

We write $\mathrm{flat}(a \otimes b) \in \mathbb{F}^{n \cdot m}$ *for the row-concatenation of $a \otimes b$.*

For our purposes, we would love to test to check if $\mathrm{flat}(a \otimes a) = b$

Definition 3.3.2. *For a field $\mathbb{F}$ and $n \in \mathbb{N}$, define the tensor test language*

$$\mathrm{TENSORQ}(\mathbb{F}, n) := \big\{(a, b) \in \mathbb{F}^n \times \mathbb{F}^{n^2} \mid b = \mathrm{flat}(a \otimes a)\big\}.$$

Theorem 3.3.3. *For every finite field $\mathbb{F}$ and $n \in \mathbb{N}$,*

$$\mathrm{TENSORQ}(\mathbb{F}, n) \in \mathrm{LPCP}\left[\varepsilon_c = 0,\ \varepsilon_s = \frac{2|\mathbb{F}| - 1}{|\mathbb{F}|^2},\ \Sigma = \mathbb{F},\ \ell = n + n^2,\ q = 3,\ r = 2n \cdot \log(|\mathbb{F}|)\right].$$

Proof. Consider the following LPCP verifier $V^{\langle a,\cdot\rangle,\langle b,\cdot\rangle}$ proceeds as follows:

1. Sample $s, t \leftarrow \mathbb{F}^n$.
2. Query a at s and at t, obtaining $\langle a, s\rangle$ and $\langle a, t\rangle$.
3. Query b at $\mathrm{flat}(s \otimes t)$, obtaining $\langle b, \mathrm{flat}(s \otimes t)\rangle$.
4. Check if $\langle b, \mathrm{flat}(s \otimes t)\rangle = \langle a, s\rangle \cdot \langle a, t\rangle$.

Completeness. Suppose $b = \mathrm{flat}(a \otimes a)$. Then $\forall s, t \in \mathbb{F}^n$,

$$\begin{aligned}\langle b,\, \mathrm{flat}(s \otimes t)\rangle = \langle \mathrm{flat}(a \otimes a),\, \mathrm{flat}(s \otimes t)\rangle &= \sum_{i,j \in [n]} a_i a_j s_i t_j \\ &= \Big(\sum_{i \in [n]} a_i s_i\Big) \cdot \Big(\sum_{j \in [n]} a_j t_j\Big) = \langle a, s\rangle \cdot \langle a, t\rangle,\end{aligned}$$

Soundness. Suppose $b \neq \mathrm{flat}(a \otimes a)$, i.e., there exists $i^*, j^* \in [n]$ such that $b_{i^* j^*} \neq a_{i^*} \cdot a_{j^*}$. For each $i \in [n]$ define the linear polynomial $p_i(t) := \sum_{j \in [n]} (b_{ij} - a_i a_j)\, t_j$. The check can be rewritten as

$$\langle b, \mathrm{flat}(s \otimes t)\rangle - \langle a, s\rangle\langle a, t\rangle = \sum_{i \in [n]} \sum_{j \in [n]} (b_{ij} - a_i a_j)\, s_i t_j = \sum_{i \in [n]} p_i(t)\, s_i.$$

Immediate application of Polynomial Identity Testing yields a lower bound of $1 - \frac{2}{|\mathbb{F}|}$ but we can do better. Since $b_{i^* j^*} \neq a_{i^*} a_{j^*}$, the polynomial p_{i^*} is nonzero, so by the Schwartz–Zippel lemma applied to t,

$$\Pr_t[p_{i^*}(t) \neq 0] \geq 1 - \tfrac{1}{|\mathbb{F}|}.$$

Conditioned on $p_{i^*}(t) \neq 0$, the expression $\sum_i p_i(t)\, s_i$ is a nonzero linear polynomial in s, so by the Schwartz–Zippel lemma applied to s,

$$\Pr_{s \leftarrow \mathbb{F}^n}\Big[\sum_i p_i(t)\, s_i \neq 0 \,\Big|\, p_{i^*}(t) \neq 0\Big] \geq 1 - \tfrac{1}{|\mathbb{F}|}.$$

Hence
$$\Pr_{s,t}[\text{verifier rejects}] \;\geq\; \left(1 - \tfrac{1}{|\mathbb{F}|}\right)^2,$$
and therefore
$$\Pr_{s,t}[\text{verifier accepts}] \;\leq\; 1 - \left(1 - \tfrac{1}{|\mathbb{F}|}\right)^2 = \frac{2|\mathbb{F}|-1}{|\mathbb{F}|^2}. \qquad \square$$

Corollary 3.3.4.
$$\mathrm{TENSORQ}(\mathbb{F}_2, n) \in \mathrm{LPCP}\left[\varepsilon_c = 0,\ \varepsilon_s = \frac{3}{4},\ \Sigma = \mathbb{F}_2,\ \ell = n + n^2,\ q = 3,\ r = 2 \cdot n\right].$$

Proof. Immediate. $\square$

3.4 Linear PCP to PCP

Lemma 3.4.1 (Chernoff bound)**.** *Let $X_1, \dots, X_N$ be independent random variables over $\{0,1\}$, and let $\mu = \mathbb{E}\left[\sum_{i=1}^N X_i\right]$. For any $\Delta > 0$ one has*
$$\Pr\left[\sum_{i=1}^N X_i \geq \mu + \Delta\right] \leq \exp(-2\Delta^2/N)$$

Proof. Write $p_i = \mathbb{E}[X_i]$, so that $\mu = \sum_{i=1}^N p_i$. We first prove the elementary exponential-moment estimate
$$\mathbb{E}[\exp(\lambda(X_i - p_i))] \leq \exp(\lambda^2/8)$$
for every $\lambda > 0$ and every i. Since X_i is $\{0,1\}$-valued,
$$\mathbb{E}[\exp(\lambda(X_i - p_i))] = (1 - p_i)\exp(-\lambda p_i) + p_i \exp(\lambda(1 - p_i)).$$
Equivalently, it is enough to show
$$\log(1 - p_i + p_i \exp(\lambda)) - p_i\lambda \leq \lambda^2/8.$$
For $p \in [0,1]$, set $f(\lambda) = \log(1 - p + p\exp(\lambda)) - p\lambda$. Then $f(0) = f'(0) = 0$, and
$$f''(\lambda) = \frac{p\exp(\lambda)}{1 - p + p\exp(\lambda)}\left(1 - \frac{p\exp(\lambda)}{1 - p + p\exp(\lambda)}\right) \leq \frac{1}{4}.$$
Hence, for $\lambda > 0$,
$$f(\lambda) = \int_0^\lambda \int_0^u f''(v)\,dv\,du \leq \int_0^\lambda \int_0^u \frac{1}{4}\,dv\,du = \lambda^2/8.$$

Let $S = \sum_{i=1}^N X_i$. By independence and the estimate above,
$$\mathbb{E}[\exp(\lambda(S - \mu))] = \prod_{i=1}^N \mathbb{E}[\exp(\lambda(X_i - p_i))] \leq \exp(N\lambda^2/8).$$
Markov's inequality gives, for every $\lambda > 0$,
$$\Pr[S \geq \mu + \Delta] = \Pr[\exp(\lambda(S - \mu)) \geq \exp(\lambda\Delta)] \leq \exp(-\lambda\Delta)\mathbb{E}[\exp(\lambda(S - \mu))]$$

Using the moment bound above, this is at most

$$\exp(-\lambda\Delta + N\lambda^2/8).$$

Taking $\lambda = 4\Delta/N$ gives

$$\Pr[S \geq \mu + \Delta] \leq \exp(-2\Delta^2/N).$$

□

Lemma 3.4.2.

$\mathrm{LPCP}[\varepsilon_c, \varepsilon_s, \Sigma = \mathbb{F}_2, \ell, q, r] \subseteq \mathrm{PCP}\left[\varepsilon_c,\ \varepsilon_s' = \max\{7/8, \varepsilon_s + 1/100\}, \Sigma' = \mathbb{F}_2,\ \ell' = 2^\ell,\ q' = 3 + 16\lceil\log(100q)\rceil q,\ r' = \right.$

Proof. Let $L \in \mathrm{LPCP}[\varepsilon_c, \varepsilon_s, \Sigma = \mathbb{F}_2, \ell, q, r]$, and let V be the LPCP verifier for L. Set $t = 8\lceil\log(100q)\rceil$, where log denotes the natural logarithm. We define a PCP verifier as follows, for any $\bar{pi} \in \mathbb{F}_2^{\mathbb{F}_2^\ell}$ and $x \in \{0,1\}^*$.

Verifier $\bar{V}^{\bar{\pi}}(x)$:

1. If $V_{\mathrm{BLR}}^{\bar{\pi}} = 0$ output 0 and halt, otherwise proceed with the next step.
2. Independently for each $i \in [t]$, sample $r_i \leftarrow \mathbb{F}_2^\ell$.
3. Simulate V by answering each linear query a with $\mathrm{plurality}(\bar{\pi}(a + r_i) - \bar{\pi}(r_i))_{i\in[t]}$.

Claim 3.4.3 (Completeness)**.** *For every $x \in L$, there exists $\bar{\pi} \in \mathbb{F}_2^{\mathbb{F}_2^\ell}$ such that* $\Pr[\bar{V}^{\bar{\pi}}(x) = 1] \geq 1 - \varepsilon_c$.

Proof. Since $x \in L$ and $L \in \mathrm{LPCP}[\varepsilon_c, \varepsilon_s, \Sigma = \mathbb{F}_2, \ell, q, r]$, let $\pi \in \mathbb{F}_2^\ell$ such that $\Pr[V^{\langle \pi, \cdot\rangle}(x) = 1] \geq 1 - \varepsilon_c$. Then we construct $\bar{\pi} \in \mathbb{F}_2^{\mathbb{F}_2^\ell}$ as follows: we define $\bar{\pi}(x) = \langle \pi, x\rangle$ for each $x \in \mathbb{F}^\ell$. Using the completeness of V_{BLR} we know that $V_{\mathrm{BLR}}^{\bar{\pi}} = 1$ with probability 1. Also, $\mathrm{plurality}(\bar{\pi}(a + r_i) - \bar{\pi}(r_i))_{i\in[t]} = \langle \pi, a\rangle$ for any query a and any r_i. We then conclude that $V^{\langle \pi, \cdot\rangle}(x) = \bar{V}^{\bar{\pi}}(x)$ for any $r_1, \dots, r_t$. Thus $\Pr[\bar{V}^{\bar{\pi}}(x) = 1] = \Pr[V^{\langle \pi, \cdot\rangle}(x) = 1] \geq 1 - \varepsilon_c$. □

Claim 3.4.4 (Soundness)**.** *For every $x \notin L$ and every $\hat{\pi} \in \mathbb{F}_2^{\mathbb{F}_2^\ell}$, one has* $\Pr[\bar{V}^{\hat{\pi}}(x) = 1] \leq \max\{7/8, \varepsilon_s + 1/100\}$.

Proof. Let $x \notin L$ and $\hat{\pi} \in \mathbb{F}_2^{\mathbb{F}_2^\ell}$. We consider two cases.

Case $\delta(\hat{\pi}, \mathbf{LIN}) \geq 1/8$. Using the soundness of V_{BLR}, we know that $\Pr[V_{\mathrm{BLR}}^{\bar{\pi}} = 0] \geq 1/8$. Since $\Pr[\bar{V}^{\hat{\pi}}(x) = 1] \leq \Pr[V_{\mathrm{BLR}}^{\bar{\pi}} = 1]$ we conclude $\Pr[\bar{V}^{\hat{\pi}}(x) = 1] \leq 1 - 1/8 = 7/8$.

Case $\delta(\hat{\pi}, \mathbf{LIN}) < 1/8$ Let $S = \{\tilde{\pi} \in \mathcal{LIN} : \delta(\hat{\pi}, \tilde{\pi} \leq \delta(\hat{\pi}, \mathrm{LIN}))\}$. Suppose for a contradiction that $|S| \geq 2$, and let $\bar{\pi}, \tilde{\pi} \in S$ be distinct: since $\delta(\hat{\pi}, \bar{\pi} \leq \delta(\hat{\pi}, \mathrm{LIN})) < 1/8$ and $\delta(\hat{\pi}, \tilde{\pi} \leq \delta(\hat{\pi}, \mathrm{LIN})) < 1/8$, by triangle inequality we have $\delta(\bar{\pi}, \tilde{\pi}) < 1/4$; since $|\mathbb{F}_2| = 2$, this contradicts the fact that any two distinct $\bar{\pi}, \tilde{\pi} \in \mathcal{LIN}$ satisfy $\delta(\bar{\pi}, \tilde{\pi}) \geq 1/2$.

Let then $\tilde{\pi}$ be the unique element of $\mathcal{LIN}$ with minimal distance to $\hat{\pi}$, and let $z \in \mathbb{F}^\ell$ the unique vector such that $\tilde{\pi}(x) = \langle z, x\rangle$ for all x. If for every query a that V asks we have $\mathrm{plurality}(\hat{\pi}(a + r_i) - \hat{\pi}(r_i))_{i\in[t]} = \tilde{\pi}(a)$, then $\bar{V}^{\hat{\pi}}(x) = V^{\langle z, \cdot\rangle}(x)$, so provided that the r_i satisfy this condition we have that $\Pr[\bar{V}^{\hat{\pi}}(x) = 1] = \Pr[V^{\langle z, \cdot\rangle}(x) = 1] \leq \varepsilon_s$. We are only left to bound the probability that for each of the q queries that V asks we have $\mathrm{plurality}(\hat{\pi}(a + r_i) - \hat{\pi}(r_i))_{i\in[t]} = \tilde{\pi}(a)$.

Note that for any a and r_i, if $\hat{\pi}(a+r_i) = \tilde{\pi}(a+r_i)$ and $\hat{\pi}(r_i) = \tilde{\pi}(r_i)$ then $\hat{\pi}(a+r_i) - \hat{\pi}(r_i) = \tilde{\pi}(a)$. Hence, the proabability over one r_i that $\hat{\pi}(a+r_i) - \hat{\pi}(r_i) \neq \tilde{\pi}(a)$ is at most $2 \cdot \delta(\hat{\pi}, \tilde{\pi}) \leq 1/4$ by a union bound. Also note that plurality$(\hat{\pi}(a+r_i) - \hat{\pi}(r_i))_{i \in [t]} \neq \tilde{\pi}(a)$ means that $\hat{\pi}(a+r_i) - \hat{\pi}(r_i) \neq \tilde{\pi}(a)$ for at least $\lceil t/2 \rceil \geq t/2$ of the i's. Let $X_i = \mathbb{1}[\hat{\pi}(a+r_i) - \hat{\pi}(r_i) \neq \tilde{\pi}(a)]$. Then $\mu = \mathbb{E}[\sum_{i \in [t]} X_i] \leq t/4$. Applying Chernoff bound with $\Delta = t/4$, the probability over the choice of $r_1, \dots, r_t$ that there are at least $t/2$ i's with $X_i = 1$ is at most $\exp(-t/8)$. Union bounding over the q queries that V asks, we have that the local correction fails with probability at most $q \cdot \exp(-t/8) \leq 1/100$ by our choice of t. □

□

Theorem 3.4.5.

$$\mathrm{QESAT}(\mathbb{F}_2, n) \in \mathrm{LPCP}[\varepsilon_c = 0, \varepsilon_s = 3/4, \Sigma = \mathbb{F}_2, \ell = n + n^2, q = 4, r = |x| + 2 * n].$$

Proof. Let $(p_1, \dots, p_m)$ be an instance of QESAT$(\mathbb{F})$ with n variables.
The LPCP verifier expects a proof $\pi = (a, b) \in \mathbb{F}^{n+n^2}$ and works as follows:

Verifier $V^{(a,b)}(p_1, \dots, p_m)$:

1. Sample $r \leftarrow \mathbb{F}^m$ and $s, t \leftarrow \mathbb{F}^n$.
2. Define
$$M := \begin{bmatrix} \mathrm{coeff}(p_1) \\ \vdots \\ \mathrm{coeff}(p_m) \end{bmatrix} \in \mathbb{F}^{m \times (n+n^2)} \quad \text{and} \quad c := \begin{bmatrix} -\mathrm{const}(p_1) \\ \vdots \\ -\mathrm{const}(p_m) \end{bmatrix} \in \mathbb{F}^m.$$
(Note: coeff(p_i) := non-constant coeffs of p_i, and const(p_i) := constant coeff of p_i)
3. *(linear equation test)* $\rightarrow$ Query (a, b) at $M^T r$ and check that $\langle a \parallel b, M^T r \rangle = \langle c, r \rangle$.
4. *(tensor test)* $\rightarrow$ Query b at flat$(s \otimes t)$, a at s and t, and check that
$$\langle b, \mathrm{flat}(s \otimes t) \rangle = \langle a, s \rangle \cdot \langle a, t \rangle.$$

Completeness: Suppose $p_1(a) = \dots = p_m(a) = 0$ and set $b := \mathrm{flat}(a \otimes a)$.
Then
$$\Pr_{s,t}[\langle b, \mathrm{flat}(s \otimes t) \rangle = \langle a, s \rangle \cdot \langle a, t \rangle] = 1$$
and
$$\Pr_r[\langle a \parallel b, M^T r \rangle = \langle c, r \rangle] = 1 \quad \left(\text{since } M \begin{bmatrix} a \\ \mathrm{flat}(a \otimes a) \end{bmatrix} = c \right).$$

Soundness: Suppose $\forall a \in \mathbb{F}^n \ \exists i \in [m]$ s.t. $p_i(a) \neq 0$. Fix any $\pi = (a, b)$.
Either:

(i) $b \neq \mathrm{flat}(a \otimes a) \implies$ tensor test passes w.p. $\leq \frac{2|\mathbb{F}|-1}{|\mathbb{F}|^2}$

OR (ii) $b = \mathrm{flat}(a \otimes a)$ and $M \begin{bmatrix} a \\ b \end{bmatrix} \neq c \implies$ linear equation test passes w.p. $\leq \frac{1}{|\mathbb{F}|}$

□